

Highlights

Beyond the Random Walk: Horizon-Weighted Temporal Fusion Transformers for Thai Agricultural Commodity Price Forecasting

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- A horizon-weighted quantile loss rebalances gradients across forecast horizons
- One pooled Temporal Fusion Transformer spans 404 Thai crop price series
- A single pooled model outperforms the Lag-1 baseline on MAE, SMAPE and RMSE
- One-year forecast error falls 37% versus naive persistence out-of-sample
- Gains are statistically significant and hold against modern foundation models

Beyond the Random Walk: Horizon-Weighted Temporal Fusion Transformers for Thai Agricultural Commodity Price Forecasting

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Abstract

Forecasting agricultural commodity prices is difficult because non-stationarity and low signal-to-noise ratios often cause high-capacity models to lose to persistence. We evaluate one global Temporal Fusion Transformer (TFT) across 404 Thai crop products at four horizons ($t + 20$, $t + 60$, $t + 120$, and $t + 250$ business days). We identify a multi-scale gradient imbalance in which far-future errors dominate joint training and introduce a **Horizon-Weighted Quantile Loss**, $w(h) = 1/h^\gamma$. A sweep of seventeen decay exponents ($\gamma \in [0, 8]$) reveals a broad accuracy plateau, and a single $\gamma = 4.5$ model is selected using the held-out 2023 validation year. On a matched 2024–2025 test set, the selected model improves every aggregate metric over Lag-1 persistence: **MAE 39.76 versus 45.21 THB (12.0% lower)**, **SMAPE 0.7% lower**, and **RMSE 15.1% lower**. The gain is concentrated at one year (**36.8% lower MAE**), whereas the model underperforms at $t + 20$ (26.51 versus 15.11 THB). Product-clustered tests identify both differences as significant ($p < 0.01$), and the per-crop win rate is **57.7%**. Results hold against seasonal-naïve, drift, and zero-shot foundation-model baselines. Validation-fitted conformal scaling gives 78–90% coverage for nominal 80% intervals. A persistence-anchored ensemble variant reaches **36.04 THB** overall without underperforming persistence at any reported horizon.

Keywords: Multi-horizon time series forecasting, Temporal Fusion Transformer, Agricultural commodity prices, Horizon-weighted loss, Quantile regression, Conformal prediction intervals

1. Introduction

Agricultural commodity markets in developing nations, particularly Thailand, are characterized by high price volatility. As a leading exporter of rice, cassava, and other crops, Thailand’s economic stability is tied to agricultural pricing. However, these prices are subject to complex domestic supply shocks, international trade policies, and market inefficiencies. Reliable price forecasting over multiple horizons is essential for risk mitigation, farm-level decision-making, and policy formulation.

Under the efficient market hypothesis, agricultural asset prices are theorized to follow random walks, rendering historical price-only structures uninformative for future horizons. Empirically, this is reflected in the difficulty that

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high-capacity models encounter when trying to outperform naive local baselines out-of-sample. This phenomenon, which we term ‘architectural saturation,’ arises because complex neural architectures (e.g., standard sequence-to-sequence Transformers, deep MLPs, and recurrent networks) possess high parameter density and capacity, allowing them to fit arbitrary functions. However, agricultural price series are characterized by low signal-to-noise ratios, non-stationarity, and frequent structural breaks. When trained on such series, high-capacity models tend to memorize transient noise and spurious historical correlations within the training sample rather than extracting persistent, structural patterns. Consequently, when evaluated on out-of-sample hold-out periods (such as 2024–2025), these over-parameterized models suffer from generalization collapse, performing significantly worse than a simple Lag-1 persistence model, which remains robust due to its zero-parameter simplicity.

To overcome the limitations of localized statistical modeling and mitigate the risk of architectural saturation, we employ cross-product representation learning using a global Temporal Fusion Transformer (TFT) architecture. Traditional local models are trained independently on each of the 404 crops, meaning they cannot share statistical strength and are highly vulnerable to localized data scarcity or reporting anomalies. In contrast, our global modeling framework pools the price histories of all 404 crops into one training dataset and fits a single set of shared model parameters. To prevent the model from conflating different crop dynamics, we supply static variables (product, category, and group identifiers) to a static covariate encoder, mapping them to low-dimensional embedding spaces. This representation learning paradigm allows the TFT to capture universal, shared features of agricultural price dynamics, such as common mean-reversion and volatility structure, general seasonal cycles, and the slow trends that related commodities share, while simultaneously conditioning its forecasts on entity-specific static embeddings to preserve the unique price-level characteristics of each individual crop. We emphasize that the case for pooling rests on this shared *functional form* rather than on correlated price levels. As Section 3.6 shows, average pairwise co-movement across the 404 crops is weak (0.18 in levels, 0.01 in daily returns), which explains why pooled models cannot outperform persistence at short horizons and why their advantage emerges only as the forecast horizon lengthens.

Importantly, the scope of our claim is bounded. At short horizons the persistence forecast is theoretically near-optimal, and our own experiments confirm that no pooled neural configuration approaches it at $t + 20$. The proposed model, a single horizon-weighted TFT, therefore concedes the near-term regime to persistence and overtakes it decisively from six months onward, where random-walk drift variance accumulates and learned macro-seasonal structure dominates. We also document a persistence-anchored ensemble variant that avoids horizon-level degradation on this test set, while not claiming that validation-fitted weights guarantee future performance.

The primary contributions of this paper are:

1. We establish a multi-scale forecasting evaluation targeting four distinct future horizons: $t + 20$ (one business month), $t + 60$ (three business months), $t + 120$ (six business months), and $t + 250$ (one business year), with model and baseline forecasts compared on identical test windows.
2. We identify and formulate the multi-scale gradient scale imbalance in joint sequence prediction, and resolve it

with a Horizon-Weighted Quantile Loss parameterized by the power-law family $w(h) = 1/h^\gamma$, mapping the full dose-response curve over $\gamma \in [0, 8]$: near-term error falls by nearly two-thirds onto a broad accuracy plateau over $\gamma \in [3, 6]$, one-year accuracy deteriorates beyond it, and raw interval quality degrades monotonically as far-horizon supervision vanishes.

3. We select the operating point ($\gamma = 4.5$) by error on a held-out validation year rather than on the test set, and report the resulting single model together with an explicit accounting of where and how much it improves on the naive baseline.
4. We show that the selected model reduces matched out-of-sample error against the Lag-1 baseline on every aggregate metric (MAE -12.0%, SMAPE -0.7%, RMSE -15.1%, and by 36.8% MAE at one year), equip it with conformally calibrated prediction intervals (78–90% empirical coverage), and document ensembling and persistence-anchored variants that trade simplicity for improved horizon-level robustness.
5. We test the horizon-level differences with a product-clustered paired t-test and Wilcoxon signed-rank test rather than reporting point estimates alone, and confirm the result against a broader reference set that includes seasonal-naive and drift baselines and a zero-shot time-series foundation model, none of which threatens the naive baseline on this dataset.

The remainder of this paper is structured as follows. Section 2 reviews related work. Section 3 details the dataset and exploratory data analysis. Section 4 describes the methodology, including the evaluation protocol, the TFT architecture, the horizon-weighted loss, and the model-selection procedure. Section 5 presents the quantitative results. Section 6 discusses model interpretability. Section 7 evaluates probabilistic calibration, and Section 8 concludes.

2. Related Work

Historically, agricultural price forecasting has been dominated by localized, parametric statistical models. The Autoregressive Integrated Moving Average (ARIMA) framework, popularized by Box and Jenkins [1], serves as a classical benchmark. These local models are fitted independently to each individual commodity series under the assumption of linear, stationary data-generating processes. While ARIMA and its seasonal variants (SARIMA) provide robust predictions in stable economic regimes, they fail to capture the high-frequency non-linear dynamics, volatility clustering, and sudden structural breaks characteristic of developing agricultural markets [2]. Furthermore, because these statistical methods are localized, they cannot share parameters across different commodities. This severely limits their predictive capacity when individual series are short, incomplete, or characterized by low signal-to-noise ratios.

To capture non-linear relationships, researchers transitioned to machine learning (ML) paradigms, such as Support Vector Regression (SVR), Random Forests, and Gradient Boosting Decision Trees (GBDT), including XGBoost and LightGBM. GBDTs have shown strong empirical performance on tabular forecasting tasks [3] due to their ability to construct non-linear decision boundaries. However, standard ML models treat sequence prediction as a static regression problem, requiring extensive manual feature engineering of rolling windows and historical lags. In a global

setting, where a single model is trained across a diverse cross-section of products, traditional GBDTs struggle to scale efficiently. They often experience severe overfitting to localized noise rather than learning generalizable sequential relationships, a precursor to architectural saturation out-of-sample.

The advent of deep learning (DL) provided a framework for end-to-end representation learning. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks [4] and Gated Recurrent Units (GRUs), introduced gated cell states to capture sequential dependencies. While RNNs address sequence learning, they suffer from vanishing gradients over long lookback windows and process inputs sequentially, which limits parallelization during training. The self-attention mechanism of the Transformer [5] resolved these bottlenecks, enabling the modeling of long-range interactions. Nevertheless, standard Transformers are highly over-parameterized and lack specialized mechanisms to handle the distinct properties of time-series data, such as static metadata and time-varying covariates, often resulting in architectural saturation where the model overfits to historical noise and fails to generalize out-of-sample.

To bridge this gap, the Temporal Fusion Transformer (TFT) was developed [6]. The TFT combines self-attention with Gated Residual Networks (GRN) and Variable Selection Networks (VSN). The VSN acts as an active information filter, selecting relevant features at each step, while the GRN provides adaptive model capacity by bypassing unnecessary layers. This architectural flexibility is critical for agricultural price forecasting, where the signal-to-noise ratio is low and overfitting is common. Furthermore, the TFT introduces interpretable self-attention, allowing researchers to inspect the exact historical steps the model prioritized when generating forecasts. Similar architectures, such as PatchTST [7] and iTransformer [8], have focused on channel-independence or patching to capture local sequences. However, they lack the native integration of static metadata (e.g., crop class, market category) and dynamic covariates in a unified framework, which is crucial for transfer learning across heterogeneous agricultural markets. Similarly, large pre-trained foundation models like Chronos [9] perform zero-shot forecasting, but they can produce spurious long-horizon trajectories when decoupled from localized, entity-specific economic context.

Finally, our system design draws on two classical strands of the forecasting literature. Forecast combination, averaging or convexly weighting heterogeneous forecasters, has been shown to improve accuracy and robustness since Bates and Granger [10], and combination strategies have dominated recent large-scale forecasting competitions such as M4 and M5 [11]. Deep ensembles [12] similarly average independently trained networks to reduce the variance intrinsic to stochastic gradient training. Our persistence-anchored ensemble applies both ideas jointly: an ensemble across loss-weighting configurations to suppress training variance, and a convex combination with the naive persistence forecast to reduce exposure to the near-term random-walk regime.

3. Data and Exploratory Analysis

Our dataset comprises daily minimum and maximum price recordings for **729 agricultural products** in Thai markets, obtained from the Ministry of Commerce (Thailand), Department of Internal Trade, via its public Open Data

portal [13], spanning January 2018 through December 2025 on a Monday–Friday business calendar (2,086 business days). Each product carries static metadata: a product identifier, a market category (retail or wholesale), and one of thirteen commodity groups ranging from fresh vegetables to wholesale rice. All statistics, tables, and figures in this section were regenerated directly from the raw data by a single reproducible analysis pass (`run_eda.py`), whose per-product profile accompanies the results as `eda_product_summary.csv`. Sections 3.1 and 3.2 document the cleaning pipeline and the definition of the modeling universe; Sections 3.3 through 3.6 then characterize the statistical structure of the data (autocorrelation, long-term trends, stationarity, and cross-product co-movement) whose properties directly motivate the design choices of Section 4.

3.1. Data Cleaning and Preprocessing

Raw agricultural price series contain reporting anomalies and data gaps. **4.60%** of the raw daily price observations (92,224 of 2,005,016) are exactly zero, and **94 series (12.89%)** are entirely empty or all-zero and are excluded outright, leaving **635 active series**. The zero values do not represent real market crashes but are artifacts of missing reports or database logging errors. The cleaning pipeline applied to every series is:

1. All zero prices are cast to missing values.
2. Outliers are removed dynamically using rolling 30-day IQR bounds ($[\text{median} - 3 \cdot \text{IQR}, \text{median} + 3 \cdot \text{IQR}]$), which adapts to structural price shifts and avoids false positives during sustained inflationary runs.
3. Missing values are imputed by Last-Observation-Carried-Forward (LOCF) forward-fill, propagating the last known valid market price.
4. The series is aligned to the standard Monday–Friday business-day skeleton to remove weekend gaps.

3.2. Product Inventory and Liquidity Filtering

Table 1 details the active and empty/all-zero series counts for each commodity group. Completeness is highly uneven. The Oil Crops & Oils group alone accounts for 44% of all empty series, while the vegetable and wholesale-rice groups are fully populated.

Beyond outright emptiness, many active listings reprice infrequently. The median active series changes price on only **2.45%** of its business days, reflecting administered prices and illiquid market segments. Because a near-constant "step-function" series is trivially forecast by any persistence-like rule, and would inflate the apparent accuracy of every model, we require economic activity for inclusion, applying a **1% volatility threshold** (a non-zero price change on at least 1% of active trading days) that retains **404 crops**, and all experiments in this paper (training, validation, and evaluation) use this single universe.

3.3. Autocorrelation Structure

We first examine the lag structure of the price series to test for random-walk characteristics. To establish that the pattern is a property of the dataset as a whole rather than of a small number of selectively chosen series, Figure 1

Table 1: Product Counts and Status by Group

Group Name	Active Products	All-Zero/Empty Products	Total Products
Meat & Poultry	98	14	112
Aquatic & Marine	39	6	45
Fresh Vegetables	89	0	89
Organic Vegetables & Fruits	44	0	44
Fruits	64	8	72
Food Crops	82	11	93
Oil Crops & Oils	95	41	136
Wholesale Rice & Bags	54	0	54
Wholesale Rice to Retailers	10	0	10
Retail Rice	9	1	10
Field Crops	40	4	44
Feed & Raw Feed Materials	11	4	15
Miscellaneous / Unknown	0	5	5
Total	635	94	729

reports the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) in levels averaged over the active series of every product group, one panel per group.

The signature is canonical unit-root behavior, and it is universal. In **every** group the PACF collapses to a single dominant spike at lag one and is negligible thereafter, meaning that once yesterday’s price is known, earlier history adds essentially nothing. The series are first-order autoregressive with a near-unit root. Tomorrow’s price is, to first order, today’s price plus noise. What distinguishes the groups is only the *rate* of ACF decay, which spans a wide and economically coherent range: storable and administered commodities exhibit minimal decay (Feed & Raw Materials 0.92 at a 40-business-day lag; wholesale and retail rice, oil crops, meat, and field crops all 0.84–0.87), whereas short-cycle perishables mean-revert fastest (Fresh Vegetables 0.39, Fruits 0.52, Aquatic & Marine 0.59), with the cross-group mean still elevated at 0.75. Storability and administered pricing, not market size, govern persistence. This spread previews the stationarity gradient quantified in Section 3.5, while the shared lag-1 structure anticipates two results of Section 5: the Lag-1 persistence baseline is exceptionally difficult to outperform at short horizons, and per-product ARIMA, which fits exactly this lag structure, collapses onto that baseline almost identically.

3.4. Long-Term Price Trends

Figure 2 tracks monthly mean prices of the six largest product groups, with each product indexed to its own 2018 average so that heterogeneous price levels share one axis.

Common macro shocks and group-specific regimes coexist. The 2022 cost shock lifts nearly every group, and oil crops approach +80% over the period with a second run-up through 2025; meat and poultry settle roughly 50% above

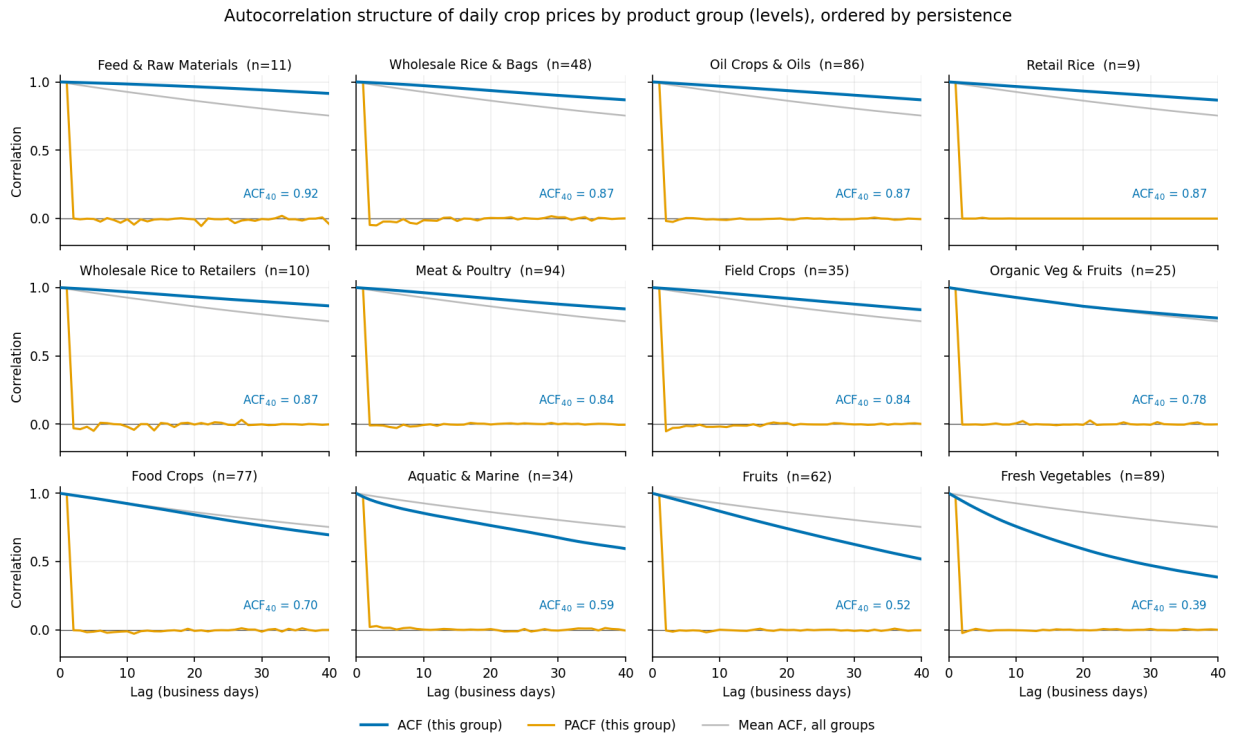


Figure 1: Mean ACF (blue) and PACF (orange) in levels over 40 lags for each of the twelve product groups with active series, ordered by persistence; the grey line repeats the cross-group mean ACF in every panel as a common reference, and n is the number of active series averaged. Every group shows the same single-spike PACF; the groups differ only in how fast the ACF decays.

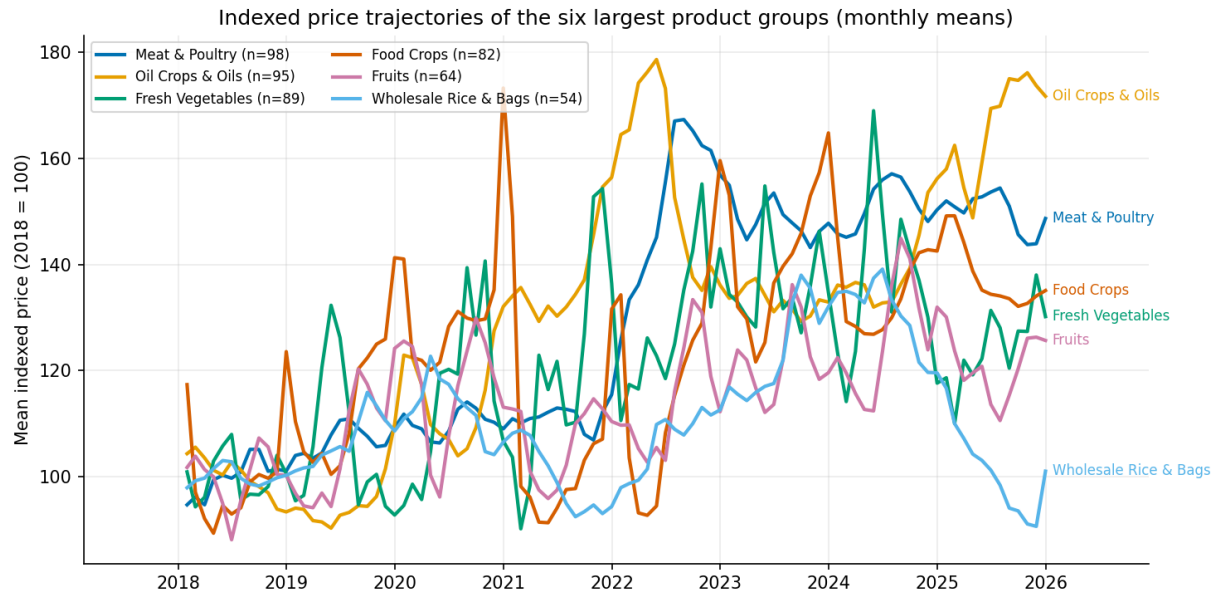


Figure 2: Indexed (2018 = 100) monthly mean price trajectories of the six largest product groups, showing heterogeneous long-term drift and volatility regimes.

their 2018 level. Wholesale rice is the structural outlier, ending 2025 at or below its 2018 level after a sustained two-year decline. A single shared dynamic cannot describe these trajectories, motivating the entity- and group-conditioned pooling of Section 4, while the shared shock structure is what a pooled model can transfer across products.

3.5. Stationarity Analysis

We formalize the unit-root evidence with Augmented Dickey-Fuller (ADF) tests in levels on every active minimum- and maximum-price series. Overall, only **30.9%** (minimum price) and **28.8%** (maximum price) of the 635 active series reject the unit root at $p < 0.05$. The dataset is predominantly non-stationary, and models must handle trend-drift and structural shifts without producing spurious regressions. Table 2 splits the rates by market category and Figure 3 by product group.

Table 2: Stationarity Profiles by Market Category (ADF, $p < 0.05$, levels)

Category	Active Series	Stationary Minimum Price (%)	Stationary Maximum Price (%)
Retail	269	35.7%	33.1%
Wholesale	366	27.3%	25.7%

Two patterns stand out. First, retail series are consistently more mean-reverting than wholesale series (35.7% vs 27.3% on minimum prices). Consumer-end price stickiness produces mean reversion, while wholesale prices track long supply trends. Second, the group-level heterogeneity is extreme. Stationarity concentrates almost entirely in short-cycle perishables (Fresh Vegetables 82%, Fruits 72%, Aquatic & Marine 59%), whose cultivation and spoilage cycles force prices back to seasonal norms, while storable and administered products (rice in every form, oil crops, livestock, feed) are almost uniformly non-stationary (0–8%). No single statistical treatment fits both regimes, which is precisely the case for conditioning a shared model on group identity.

3.6. Cross-Product Co-Movement

Finally, we quantify market integration across the modelling universe. Over all 81,406 product pairs in the 404-crop universe, co-movement is **weak on average**. The mean pairwise correlation of price levels is only **0.180** (median 0.176), and just **14.3%** of pairs exceed 0.5. In daily *returns* it effectively vanishes, with a mean of **0.012** across all pairs and **-0.001** between products of different groups. Integration is concentrated rather than broad. Within a commodity group, level correlation rises to 0.357 (returns 0.091), against 0.152 (returns -0.001) across groups.

Figure 4 makes the concentration visible by zooming in on its extreme, the 40 most liquid series, which are dominated by internationally traded rubber, palm, cassava, and grain contracts.

Within these families the correlation is near-mechanical (mean **0.940** in levels). The rubber block is a single USS3/RSS3 sheet grade quoted at three provinces and two market levels, and the palm block is crude, RBD, olein and stearin variants of one oil complex plus fresh fruit bunches priced by province. Even here, cross-family correlation falls to 0.481 in levels and **0.037** in returns. Grains and rubber share a slow trend, not a daily shock.

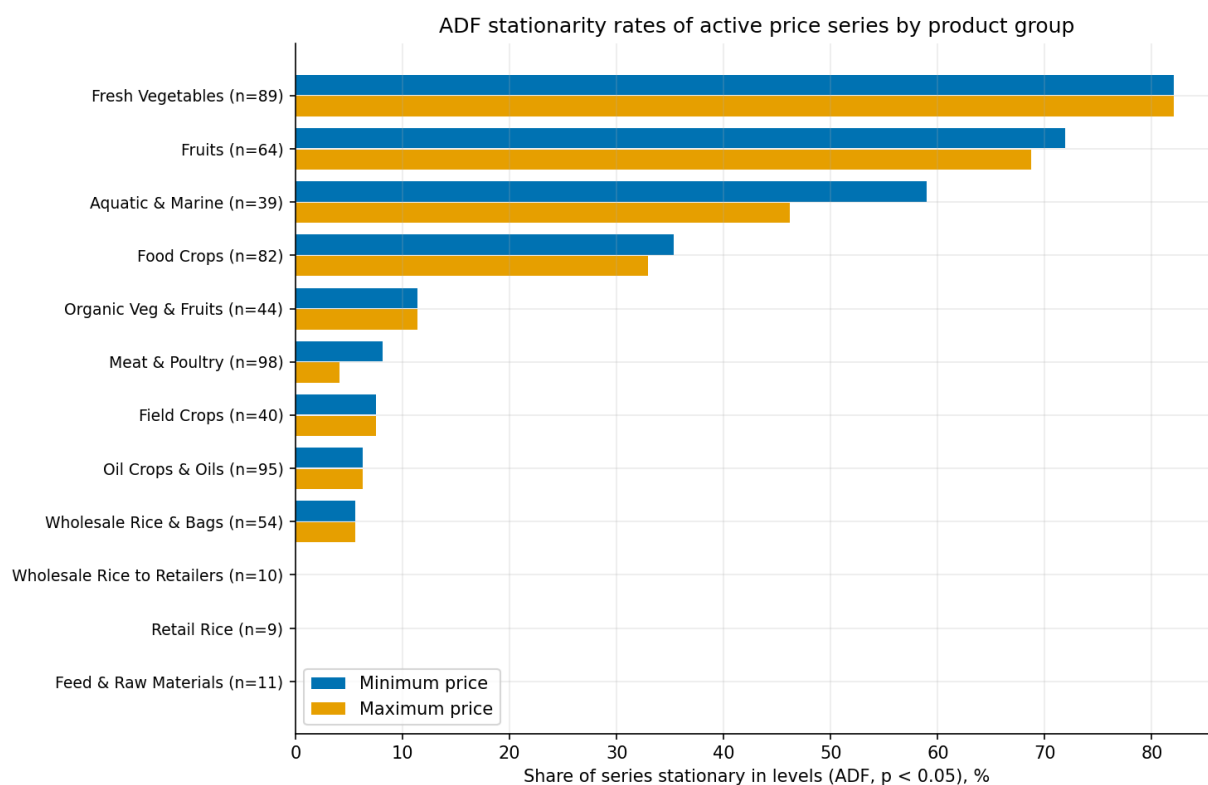


Figure 3: ADF stationarity rates for active minimum and maximum price series by product group. Thai group names are translated to concise English labels for publication readability; n denotes the number of active series.

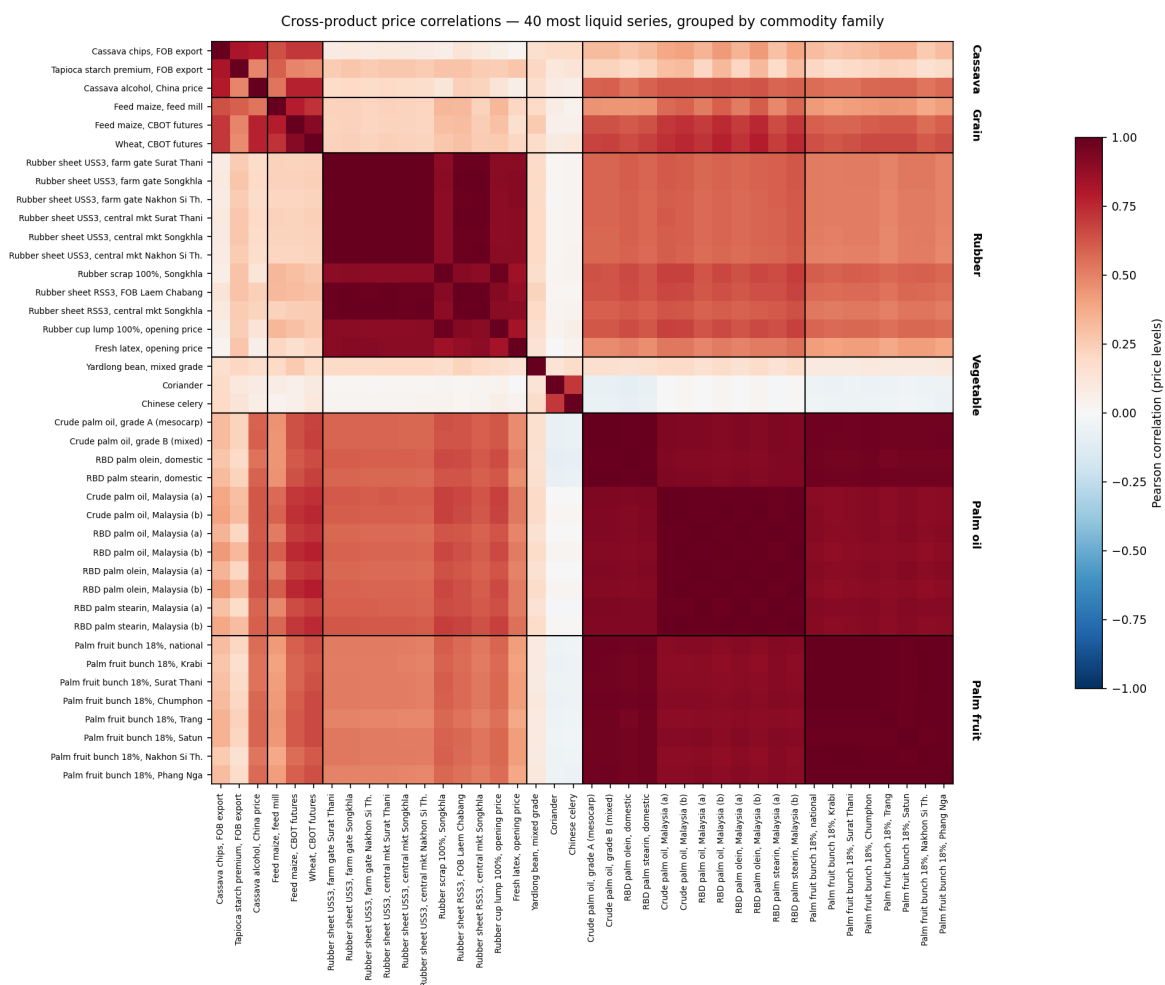


Figure 4: Pairwise Pearson correlation of price levels for the 40 most liquid series, ordered by commodity family (families annotated by the authors from the product names; the dataset's own grouping assigns rubber, cassava, and grain alike to "Field Crops"). This panel deliberately depicts the most integrated segment of the dataset; the near-unit blocks are largely one commodity priced at several grades, provinces, or delivery points, and it is not representative of the 404-crop universe, whose average correlation is 0.180.

Two conclusions follow, and together they frame the rest of the paper. First, the pooling argument cannot rest on correlated prices. At 0.18 average level correlation and approximately 0.01 return correlation, there is essentially no cross-sectional signal to exploit, which is precisely why no pooled model in Section 5 improves on persistence at short horizons. What a global model can share instead is the *functional form* of the dynamics (the mean-reversion, seasonal, and volatility structure common to agricultural prices) plus direct transfer among the related series that do co-move (0.36 within group, 0.94 among near-duplicates). Second, what integration exists lives entirely in slow trends rather than daily shocks, which is why the pooled advantage in Section 5 appears only as the horizon lengthens.

In summary, the exploratory analysis yields four modeling prescriptions: (i) treat zero prices and outliers before any model sees the data; (ii) filter economically inactive listings, since near-constant series inflate the apparent accuracy of every forecaster; (iii) expect persistence to dominate short horizons, since the data contain neither the autocorrelation structure nor the cross-sectional co-movement needed to outperform it there, and design the objective accordingly rather than against it; and (iv) pool across products for shared dynamics and long-horizon trend structure, while conditioning on product and group identity to absorb the extreme heterogeneity in drift, mean reversion, and price level.

4. Methodology

4.1. Forecasting Horizons and Evaluation Metrics

We establish a multi-scale forecasting horizon aligned with agricultural decision cycles. Given the input sequence, models forecast at specific future steps:

- $t + 20$: One business month / 20 business days.
- $t + 60$: Three business months / 60 business days.
- $t + 120$: Six business months / 120 business days.
- $t + 250$: One business year / 250 business days.

Performance is measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Symmetric Mean Absolute Percentage Error (SMAPE), calculated as:

$$\begin{aligned} \text{MAE} &= \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \\ \text{RMSE} &= \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \\ \text{SMAPE} &= \frac{100\%}{N} \sum_{i=1}^N \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2} \end{aligned}$$

where y_i represents the actual price and $\hat{y}_i$ represents the predicted price. For every quantile-regression TFT in this study (Section 4.4, item 5), $\hat{y}_i$ is the network’s median (0.5 quantile) output, $\hat{y}_i^{med}$ in the notation of Section 4.7; the 0.1 and 0.9 quantile heads contribute only to the prediction intervals of Section 7 and never to the point-forecast metrics reported in Tables 5 through 8. For the non-quantile reference models of Section 4.3, $\hat{y}_i$ is each model’s single point output.

4.2. Evaluation Protocol and Matched-Sample Comparison

All results follow a strict out-of-time protocol, summarized in Table 3. Model parameters are learned on labels through the end of 2022 over the pooled 404-crop dataset; the 2023 calendar year serves as the validation period for early stopping and for fitting all post-hoc calibration parameters; the 2024–2025 period is reserved exclusively for final evaluation. For every forecast origin in the test period, the model and the Lag-1 persistence baseline are evaluated on the *identical* window, with the persistence forecast defined as the final observed value of that window’s encoder segment. This matched-sample design eliminates a subtle but serious failure mode in which model and baseline metrics are computed over different origin distributions and become incomparable. Each of the four reporting horizons is evaluated over 110,292 test windows (273 origins for each of the 404 crops); because every crop contributes an identical number of windows, the pooled average coincides exactly with the per-product macro-average. Validation-side fitting of calibration (and blend-variant) parameters uses, per horizon, every window whose label at that horizon falls within 2023 (12,120 windows at $t + 20$ growing to 105,040 at $t + 250$), so that even the longest horizon is fitted on a full year of labels rather than a small number of early-year origins.

Table 3: Chronological Data Partition and Its Role in the Protocol. Validation labels serve early stopping, decay-exponent selection, interval calibration, and blend-weight fitting; no test-period information enters any fitted quantity

Partition	Period	Labels used for	Windows per horizon
Training	2018–2022	Parameter estimation (404 crops, pooled)	n/a
Validation	2023	Early stopping; selection; calibration	12,120–105,040
Test	2024–2025	Final evaluation only	110,292

4.3. Baseline and Reference Models

We compare the proposed model against a suite of localized statistical benchmarks and global machine learning and deep learning references. The suite is constructed so that the comparisons of Section 5 isolate four distinct questions: whether any classical local rule, naive or fitted, is hard to beat (persistence, seasonal-naive, drift, and ARIMA), whether pooled learning over engineered features adds value without any sequence modeling (LightGBM), whether increasing architectural capacity adds value under identical inputs (MLP, gated recurrent networks, and a standard Transformer), and whether large-scale pretraining on heterogeneous time series substitutes for in-domain training (a

zero-shot foundation model). All learned references share the conditions described below, so that differences in Table 5 are attributable to model class rather than to data, features, or evaluation.

Common feature representation. Every learned reference model consumes the same engineered, leakage-safe representation of 51 price-derived and calendar predictors per forecast origin: twelve price lags spanning 1 to 250 business days; six differences and six returns over windows of 1 to 250 days; fourteen rolling statistics (means, standard deviations, medians, extrema, range, and coefficient of variation over 7- to 120-day windows); four repricing-activity counters (days since the last price change and change counts over 20, 60, and 250 days); and nine calendar coordinates (month, weekday, day of year, and their sine/cosine encodings). Every feature is computed strictly from prices available at or before the forecast origin, so no feature contains label information. Global models additionally receive integer-encoded product, category, and group identifiers.

Common training and evaluation conditions. Reference models are fitted on all data through the end of 2023, the union of the proposed model’s training (2018–2022) and validation (2023) periods, so every competitor receives at least as much historical data as the TFT. Evaluation follows the matched-window protocol of Section 4.2 on identical 2024–2025 test windows. Multi-horizon prediction follows the strategy natural to each model class: recursive forecasting for ARIMA, direct forecasting (one independently trained model per horizon) for LightGBM, and joint multi-output regression for the neural networks. For the neural models, continuous inputs are standardized and targets are standardized during training, with predictions transformed back to price units for evaluation. As a final safeguard, any non-finite prediction is replaced by the persistence value for that window, a conservative rule that can only improve a competitor’s measured accuracy. The evaluation pipeline also covers local per-product variants of each learned model and a random-forest reference in both paradigms; none of these outperforms the naive baseline overall, and none alters any conclusion of Section 5, so Table 5 reports the global variants consistent with the pooled-modeling focus of this study. To confirm that no reference class is handicapped by its configuration, the Supplementary Material reports a validation-based hyperparameter search for every learned reference class under an equal light budget (ten candidate configurations per class, matching the budget of the TFT’s own architecture search, with the published configuration always among the candidates). No validation-selected configuration outperforms the persistence baseline overall or approaches the proposed model, so the comparisons of Section 5 are insensitive to reference-model tuning at this budget.

1. **Lag-1 Persistence Baseline:** The Lag-1 persistence baseline issues the naive forecast of a random walk, carrying the most recently observed price forward to every horizon:

$$\hat{y}_{t+h} = y_t, \quad h \in \{20, 60, 120, 250\}$$

Its status as a benchmark is theoretically grounded. If prices follow a martingale, as the efficient market hypothesis implies for speculative assets, the conditional expectation of any future price equals the current price, and persistence is the minimum mean-squared-error forecast that no fitted model can systematically outperform. The autocorrelation evidence of Section 3.3 places these series close to that regime, making this zero-parameter

rule the single most demanding benchmark in the study. It also serves as the fallback forecast throughout the pipeline.

2. **Seasonal-Naive and Drift:** Two further zero-parameter classical rules that a horizon-weighted result must survive. Seasonal-naive repeats the price observed exactly one full decoder cycle (250 business days, approximately one year) before the target, $\hat{y}_{t+h} = y_{t+h-250}$, always a known quantity at the forecast origin since $h \leq 250$; it targets the annual crop cycles documented in Section 3.4 directly, at the cost of ignoring everything between. Drift extrapolates the straight line through the first and last observed prices of each series' history up to the origin, $\hat{y}_{t+h} = y_t + h \cdot (y_t - y_1)/t$, the classical alternative to persistence when a trend is suspected [14]. Both are included specifically because they are strong long-horizon competitors on seasonal or trending series, and Section 5.1 confirms neither threatens Lag-1 persistence on this dataset.
3. **Local ARIMA:** An ARIMA(1, 1, 0) specification is fitted independently to each of the 404 crop price series by state-space maximum likelihood. The order is chosen to mirror the empirical lag structure documented in Section 3.3, a unit root plus a single partial-autocorrelation spike at lag one:

$$\Delta y_t = c + \phi_1 \Delta y_{t-1} + \epsilon_t$$

where Δ is the first-difference operator, ϕ_1 the autoregressive coefficient, and ϵ_t white noise. Multi-step forecasts are generated recursively from every test origin through the Kalman filter, and any series whose estimation fails to converge falls back to persistence. The forecast function makes the near-identity with persistence in Table 5 interpretable rather than surprising: iterating the recursion gives $\hat{y}_{t+h} = y_t + \phi_1 \frac{1-\phi_1^h}{1-\phi_1} \Delta y_t$ (for $c = 0$), which flattens within a few steps at a level differing from y_t by at most $\frac{\phi_1}{1-\phi_1} \Delta y_t$. Because the levels are near-unit-root first-order autoregressive (Section 3.3), the differenced series is close to white noise, the estimated ϕ_1 is small, and the forecast is numerically indistinguishable from the naive rule at every reporting horizon. Its linear assumptions and inability to share parameters across series further limit its capacity to model non-linear price transmission.

4. **Global MLP:** A feedforward network trained on the pooled cross-section of all 404 crops. Entity information enters through learned embeddings (a 16-dimensional product embedding, 8-dimensional category and group embeddings, and 4-dimensional month and weekday embeddings), which are concatenated with the 49 standardized continuous features into a single input vector. Two fully connected hidden layers of 128 and 64 ReLU units, each followed by dropout of 0.47, map this vector to a joint multi-output regression head covering all forecast horizons simultaneously. Training minimizes mean squared error with Adam (learning rate 3.4×10^{-3} , batch size 4,096, five epochs). The MLP isolates whether pooled non-linear regression on the engineered features, with no sequential inductive bias whatsoever, can extract structure that the persistence rule misses.
5. **Global LightGBM:** LightGBM [3] is an optimized gradient-boosted decision tree (GBDT) framework. We use direct forecasting with one independently trained booster per horizon $h \in \{20, 60, 120, 250\}$, which avoids the error accumulation of recursive strategies at long horizons. Each booster is an additive ensemble of 100

leaf-wise trees trained with an L2 objective (learning rate 0.05, up to 2,633 leaves at a maximum depth of 8, a minimum of 700 observations per leaf, feature subsampling of 0.99, and bagging fraction 0.74 applied every third iteration). Product, category, group, month, and weekday enter through LightGBM’s native categorical-split handling rather than through embeddings. GBDT ensembles are consistently strong learners on engineered tabular features, and Section 5.1 confirms this reference as the strongest learned alternative to the proposed model.

6. **Global LSTM and GRU:** Single-layer gated recurrent networks with a 16-dimensional hidden state, trained globally across all commodities with the same entity embeddings and standardized continuous inputs as the MLP and the same joint multi-output head, optimized with Adam (learning rate 0.01, batch size 4,096, three epochs). In this design the sequential history enters through the engineered lag and rolling-window features, so the gated cells act as learned non-linear filters over that representation. Unlike the TFT, these architectures possess neither self-attention nor variable selection networks, forcing them to process all input variables uniformly. Table 5 reports the LSTM; the GRU variant performs comparably (53.82 THB overall MAE against the LSTM’s 59.59) and, like every omitted reference variant, does not outperform the naive baseline overall.
7. **Standard Transformer:** The standard Transformer reference [5] applies an encoder of two multi-head self-attention layers (four heads, model width $d_{model} = 32$, feed-forward width 128, dropout 0.1) to a six-token input in which the projected continuous feature vector and the five entity and calendar embeddings each constitute one token:

$$\text{head}_h = \text{Softmax} \left(\frac{\mathbf{Q}_h \mathbf{K}_h^T}{\sqrt{d_k}} \right) \mathbf{V}_h$$

followed by position-wise feed-forward layers, residual connections, and layer normalization; the flattened encoder output feeds the same joint multi-output head as the other neural references (Adam, learning rate 3×10^{-3} , batch size 4,096, five epochs). This model serves as the unstructured high-capacity control. It possesses full attention capacity but none of the time-series-specific inductive biases of Section 4.4, and it is prone to overfitting and architectural saturation on noisy series due to the lack of variable selection gating and static metadata conditioning, a prediction that Section 5.1 confirms empirically.

8. **Zero-Shot Foundation Model:** Chronos-Bolt-Base [9] is a pretrained probabilistic time-series foundation model, included to test whether large-scale pretraining on heterogeneous public time series substitutes for the in-domain training every other reference receives. It is evaluated zero-shot: no fine-tuning, no exposure to this dataset during pretraining, and the same test windows as every other row of Table 5. Each forecast conditions on up to 512 business days of that product’s own price history immediately preceding the origin (the model’s practical context limit; encoder histories shorter than 512 days use whatever history is available), predicts 250 steps ahead, and reports the median of the predictive distribution at each reporting horizon.

Significance testing. For the two head-to-head comparisons the paper makes most of (the selected model against the matched baseline in Section 5.3, and against LightGBM implicitly via Table 5), we additionally test whether the

observed MAE differences are distinguishable from zero. A standard time-series Diebold-Mariano test [15] is not valid here: forecasts from consecutive daily origins at a fixed horizon overlap by construction, and at $h = 250$ the 273 test-period origins per crop contain close to one effective independent observation, a degeneracy no long-run-variance correction can repair. We instead cluster by product. For every crop we collapse its per-window absolute-error differentials (selected model minus baseline) to one mean value, yielding one approximately independent observation per crop; the near-zero cross-product return correlation documented in Section 3.6 (mean 0.012) supports treating these 404 values as independent for testing purposes. We report a two-sided paired t-test and a Wilcoxon signed-rank test on this cross-section of per-product mean differentials, per horizon and overall.

4.4. Temporal Fusion Transformer Architecture

The Temporal Fusion Transformer (TFT) architecture integrates static metadata and multi-scale sequential covariates using specialized neural blocks. The network is optimized end-to-end to generate multi-horizon probabilistic forecasts. All inputs to the TFT in our experiments are strictly price-derived (the historical target series) plus deterministic calendar covariates and categorical metadata (product, category, and market group IDs).

Information flows through the network as follows. Static metadata is first mapped to entity embeddings and, through four dedicated gated residual networks, to context vectors that condition every downstream block (Section 4.5). At each time step, a variable selection network weighs the available inputs; the selected representation passes through an LSTM encoder over the 30-day lookback window and an LSTM decoder over the 250-step forecast window, both initialized from the static context. Gated skip connections and an interpretable multi-head self-attention layer then enrich the decoder states with long-range dependencies, and linear quantile heads emit the 10th, 50th, and 90th percentile forecasts at every decoder step, trained jointly under the horizon-weighted quantile loss of Section 4.6. The individual blocks are detailed below, and Table 4 consolidates the full configuration.

4.4.1. Gated Residual Networks (GRN) and GLU

To adaptively allocate model capacity and filter out noise from weak predictors, the TFT uses Gated Residual Networks (GRN) as its primary building blocks. Given an input vector $\mathbf{a} \in \mathbb{R}^{d_m}$ and an optional static context vector $\mathbf{c} \in \mathbb{R}^{d_{context}}$, the GRN is formulated as:

$$\text{GRN}_{d_{out}}(\mathbf{a}, \mathbf{c}) = \text{LayerNorm}(\mathbf{a} + \text{GLU}_{d_{out}}(\eta_1))$$

where the gating mechanism is governed by a Gated Linear Unit (GLU):

$$\text{GLU}_{d_{out}}(\gamma) = \sigma(\mathbf{W}_4\gamma + \mathbf{b}_4) \odot (\mathbf{W}_5\gamma + \mathbf{b}_5)$$

and the intermediate activations are defined as:

$$\eta_1 = \mathbf{W}_1\eta_2 + \mathbf{b}_1$$

$$\eta_2 = \text{ELU}(\mathbf{W}_2 \mathbf{a} + \mathbf{W}_3 \mathbf{c} + \mathbf{b}_2)$$

Here, ELU is the Exponential Linear Unit activation, σ is the sigmoid function, $\odot$ represents the Hadamard product, and $\mathbf{W}_i, \mathbf{b}_i$ are learnable weights and biases. When a feature contains mostly noise, the GLU's sigmoid gate suppresses the non-linear path, reverting the GRN to a simple linear mapping, which directly mitigates architectural saturation.

4.4.2. Variable Selection Networks (VSN)

At each time step, a VSN acts as an active information filter, dynamically identifying the most relevant features. For a set of M variables $\mathbf{x}_t = [\mathbf{x}_t^{(1)}, \dots, \mathbf{x}_t^{(M)}]^T$, the network computes variable selection weights $\mathbf{v}_t \in \mathbb{R}^M$:

$$\mathbf{v}_t = \text{Softmax}(\text{GRN}_v(\mathbf{g}_t, \mathbf{c}))$$

where $\mathbf{g}_t = [(\tilde{\mathbf{x}}_t^{(1)})^T, \dots, (\tilde{\mathbf{x}}_t^{(M)})^T]^T$ is the concatenated representation of all variables projected into d_{model} -dimensional spaces via independent GRNs:

$$\tilde{\mathbf{x}}_t^{(j)} = \text{GRN}_{x,j}(\mathbf{x}_t^{(j)})$$

The final selected feature vector is a weighted sum:

$$\tilde{\mathbf{x}}_t = \sum_{j=1}^M v_t^{(j)} \tilde{\mathbf{x}}_t^{(j)}$$

This allows the model to isolate and prioritize high-signal features (like immediate lags) and completely ignore noisy inputs.

4.4.3. LSTM Sequence Encoder/Decoder

The selected features are passed into a sequence-to-sequence layer for local context processing. An LSTM encoder processes the historical lookback window ($k = 30$ business days), and an LSTM decoder processes the full 250-step forecast window from which the reporting horizons $h \in \{20, 60, 120, 250\}$ are read. The recurrent states are initialized from static context vectors:

$$\mathbf{h}_0 = \mathbf{c}_h, \quad \mathbf{c}_0 = \mathbf{c}_c$$

where $\mathbf{h}_0$ is the initial hidden state, $\mathbf{c}_0$ is the initial cell state, and $\mathbf{c}_h, \mathbf{c}_c$ are GRN outputs of the full static covariate vector (Section 4.5), allowing the recurrent layer to adapt its dynamics to the specific commodity. The LSTM cell updates at time step t are governed by the standard gating mechanisms:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f)$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t)$$

where $\mathbf{f}_t, \mathbf{i}_t, \mathbf{o}_t$ represent the forget, input, and output gates, $\mathbf{c}_t$ is the cell state, and $\mathbf{h}_t$ is the hidden state.

4.4.4. Interpretable Multi-Head Self-Attention

To capture long-term sequential dependencies, the TFT uses a modified multi-head self-attention layer. Standard multi-head attention projects queries, keys, and values independently per head, making the resulting attention maps uninterpretable. The TFT resolves this by sharing query and key weights across all heads while projecting values independently:

$$\text{InterpretableMultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \tilde{\mathbf{A}}(\mathbf{Q}, \mathbf{K})\mathbf{V}\mathbf{W}_o$$

$$\tilde{\mathbf{A}}(\mathbf{Q}, \mathbf{K}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_{\text{attn}}}}\right)$$

where $\mathbf{Q} = \mathbf{\Theta}\mathbf{W}_Q$ and $\mathbf{K} = \mathbf{\Theta}\mathbf{W}_K$ are shared query and key projections. This generates a single, unified attention matrix $\tilde{\mathbf{A}}$ that represents the exact historical steps the model prioritized when generating forecasts.

4.4.5. Quantile Loss Function

To generate probabilistic intervals rather than simple point predictions, the model is optimized using joint quantile regression. For target quantiles $\mathcal{Q} = \{0.1, 0.5, 0.9\}$ (defining an 80% prediction interval around the median), the Quantile Loss is formulated as:

$$\mathcal{L}(\Omega) = \sum_{y \in \Omega} \sum_{q \in \mathcal{Q}} \sum_{t=1}^T \rho_q(y_t - \hat{y}_{t,q})$$

where $\rho_q(e)$ is the pinball loss function:

$$\rho_q(e) = e(q - \mathbb{I}_{\{e < 0\}})$$

4.4.6. Training Configuration and Hyperparameters

Table 4 consolidates the complete configuration, which is identical for every TFT in this study; the decay exponent γ of the loss function (Section 4.6) is the only quantity varied across experiments, so that any performance difference in Section 5.2 is attributable to the loss geometry alone. All experiments are conducted globally over the pooled crop dataset with a fixed random initialization per run and are accelerated with PyTorch and CUDA on dedicated GPU hardware (NVIDIA RTX series). The full training and evaluation pipeline, including per-run artifacts (metrics, per-sample predictions, and model checkpoints), is reproducible from the accompanying code.

4.5. Cross-Product Representation Learning and Static Covariate Encoders

A primary challenge of global forecasting across heterogeneous price series is capturing product-specific characteristics without parameter bloat. The TFT resolves this through cross-product representation learning. First, categorical metadata variables representing the product ($p \in \mathcal{P}$), category ($c \in \mathcal{C}$), and group ($g \in \mathcal{G}$) are mapped to

Table 4: Temporal Fusion Transformer Configuration (identical across all experiments)

Component	Setting
Encoder (lookback) length	30 business days
Decoder (forecast) length	250 business days; reporting horizons $t + 20$, $t + 60$, $t + 120$, $t + 250$
Hidden state size	64
Attention heads	4
Continuous-variable encoding size	8
Dropout	0.1
Output quantiles	{0.1, 0.5, 0.9}
Target normalization	Per-product group normalizer, softplus transformation
Loss function	Horizon-Weighted Quantile Loss, $w(h) = 1/h^\gamma$ (Section 4.6)
Optimizer	Adam, learning rate 0.01, reduce-on-plateau schedule (patience 4)
Batch size	64 (training), 128 (inference)
Gradient clipping	0.1
Epoch budget	5, with early stopping on validation loss (patience 5, min. delta 10^{-4})

low-dimensional continuous vector embeddings:

$$\mathbf{e}_p = \mathbf{W}_p \mathbf{x}_p, \quad \mathbf{e}_c = \mathbf{W}_c \mathbf{x}_c, \quad \mathbf{e}_g = \mathbf{W}_g \mathbf{x}_g$$

where $\mathbf{x}_p, \mathbf{x}_c, \mathbf{x}_g$ are one-hot encoded vectors, and $\mathbf{W}_p \in \mathbb{R}^{d \times |\mathcal{P}|}$, $\mathbf{W}_c \in \mathbb{R}^{d \times |\mathcal{C}|}$, $\mathbf{W}_g \in \mathbb{R}^{d \times |\mathcal{G}|}$ are learnable embedding matrices.

These static embeddings are concatenated to form the static covariate representation:

$$\mathbf{s} = [\mathbf{e}_p^T, \mathbf{e}_c^T, \mathbf{e}_g^T]^T$$

The static vector $\mathbf{s}$ is then passed through four independent Gated Residual Networks to generate specialized static context vectors:

$$\mathbf{c}_s = \text{GRN}_s(\mathbf{s}), \quad \mathbf{c}_e = \text{GRN}_e(\mathbf{s}), \quad \mathbf{c}_c = \text{GRN}_c(\mathbf{s}), \quad \mathbf{c}_h = \text{GRN}_h(\mathbf{s})$$

These context vectors integrate the static metadata directly into the sequence-to-sequence networks:

- $\mathbf{c}_s$ conditions the variable selection network (VSN) for input features.
- $\mathbf{c}_c$ and $\mathbf{c}_h$ initialize the cell state ($\mathbf{c}_0$) and hidden state ($\mathbf{h}_0$) of the LSTM sequence encoder and decoder:

$$\mathbf{c}_0 = \mathbf{c}_c, \quad \mathbf{h}_0 = \mathbf{c}_h$$

- $\mathbf{c}_e$ modulates the intermediate sequential representations before the self-attention layer.

This static covariate encoding structure allows the global model to construct a shared representational space for general price dynamics (e.g. macro-economic cycles) while utilizing static context vectors to customize the model’s behavior for each individual crop.

4.6. Multi-Horizon Gradient Conflict and the Horizon-Weighted Quantile Loss

In a standard joint multi-step sequence loss, the gradient magnitudes are dominated by the far-future steps, where prediction absolute errors are naturally several times larger than near-future errors (in our test data, baseline MAE grows from roughly 15 THB at $t + 20$ to 84 THB at $t + 250$). This causes backpropagation updates to prioritize long-term seasonal patterns at the expense of near-term price transmission dynamics, leading to sub-optimal near-term convergence.

To balance the gradients across lookahead horizons, we apply a step-dependent scale weight $w(h)$ to the quantile loss at every decoder step:

$$\mathcal{L}_{HW-Quantile} = \sum_{h=1}^H w(h) \cdot \mathcal{L}_{Quantile}(y_{pred,t+h}, y_{true,t+h})$$

We parameterize the loss decay weighting function using a 1-parameter power-law family:

$$w(h) = \frac{1}{h^\gamma}$$

where $\gamma \geq 0$ is the decay exponent that controls the rate at which gradients from future steps are discounted during optimization.

We adopt a power-law family, rather than an exponential or other decay schedule, for three reasons. First, a single-exponent power law is scale-free. Rescaling the horizon by a constant factor rescales the weight by a constant factor, independent of the horizon itself. The same γ therefore governs the relative down-weighting between $t + 20$ and $t + 60$ as it does between $t + 120$ and $t + 250$, even though our four reporting horizons span more than an order of magnitude. An exponential family $w(h) = e^{-\lambda h}$ lacks this property. It introduces a characteristic decay horizon $1/\lambda$ that must be fixed a priori, and a single λ cannot suit near-term and one-year dynamics simultaneously without either flattening the near horizons or extinguishing the far ones. Second, a single-exponent power-law modulation of a per-instance loss term has direct precedent in the deep learning literature. Focal loss [16] reweights the cross-entropy loss by $(1 - p_t)^\gamma$ to down-weight easy examples in dense object detection, using one exponent to interpolate continuously between the unweighted loss ($\gamma = 0$) and increasingly aggressive focusing. Our $w(h) = 1/h^\gamma$ plays an analogous role along the horizon axis rather than the class-confidence axis, with $\gamma = 0$ recovering the unweighted multi-horizon loss and increasing γ continuously sharpening the network’s focus toward near-term steps. We prefer this fixed, closed-form weighting to adaptive multi-task balancing schemes such as GradNorm [17], which learn per-task weights jointly with the network, because a weight fixed before training begins can be swept exhaustively as in Section 5.2 and introduces no additional optimization dynamics of its own. Third, and specific to price forecasting, the exponent has a direct interpretation under random-walk dynamics. As we derive formally in Section 5.3, the persistence forecast error of

a random walk has variance $\text{Var}(y_{t+h} - y_t) = h\sigma^2$ [1], so its standard deviation, and with it the natural scale of the per-horizon loss, grows as $O(\sqrt{h})$. The $\gamma = 0.5$ configuration in our sweep is therefore not an arbitrary grid point. It is the exponent that exactly offsets this theoretical error growth, and the remaining configurations probe departures from that baseline in both directions.

Under this formulation, we systematically evaluate seventeen decay exponents spanning $\gamma \in \{0.0, 0.5, \dots, 8.0\}$ in half-unit steps, ranging from the standard unweighted objective ($\gamma = 0$, $w(h) = 1$) through square-root, linear, and inverse-square decays out to extreme prioritization of near-term gradients (at $\gamma = 8$ the weight on the 250th step is of order 10^{-20} , so effectively only the first few forecast steps carry training signal). The comparative shapes of representative decay curves over the 250-day forecast window are plotted in Figure 5.

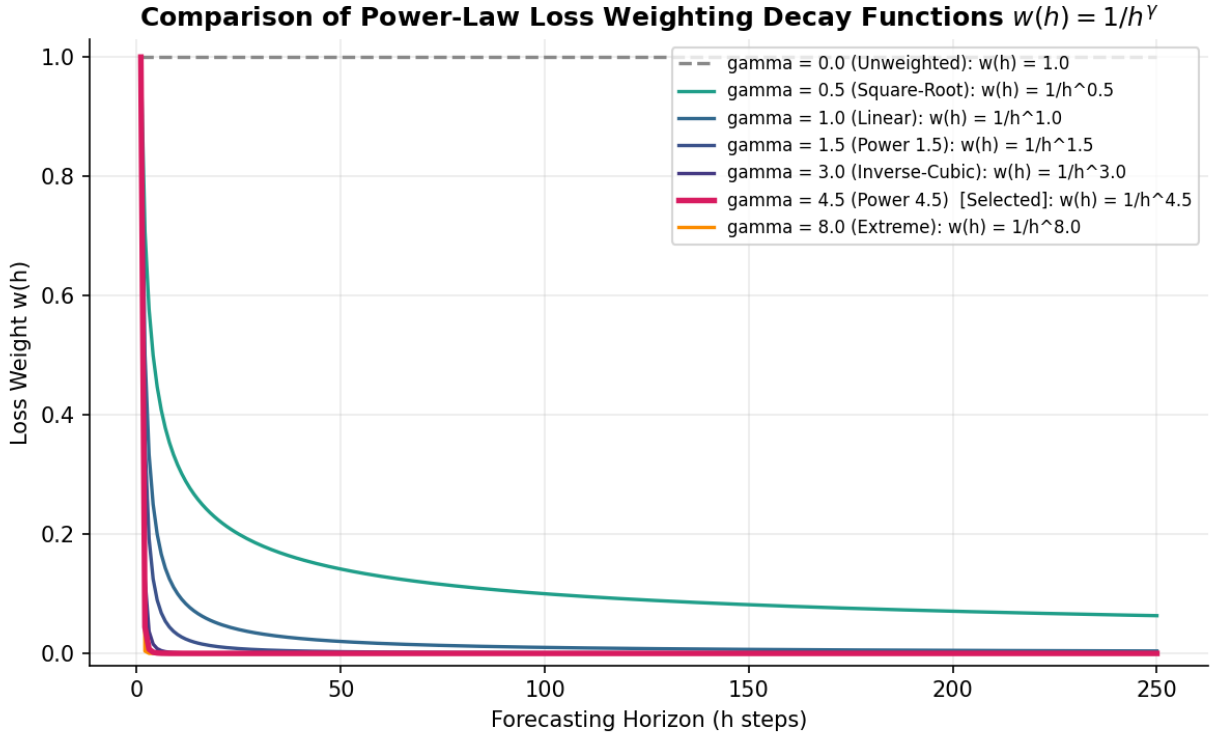


Figure 5: Comparison of the unweighted baseline and alternative scale-weighting decay functions used to balance backpropagation gradients across horizons.

4.7. Replication, Model Selection, and Interval Calibration

The horizon-weighted loss addresses gradient conflict *within* a single network, but repeated training of identical TFT configurations on this low signal-to-noise data produces substantial run-to-run dispersion in test error (independent retraining of the same γ shifted $t + 20$ MAE by up to 18 THB in our experiments), so conclusions attached to any single trained network are fragile. Our selection protocol therefore rests on replication:

1. Model selection on validation. The selected configuration is the decay exponent with the lowest error on the 2023 validation year; among the swept exponents, $\gamma = 4.5$ attains the best validation MAE and is selected. No test-period information enters the choice. Because pooled TFTs of this kind show non-trivial run-to-run variance on this low signal-to-noise data, the exponent was retrained under independent initializations to confirm its stability before selection; Section 5.2 discusses the resulting variance.

2. Conformal interval calibration. Raw quantile intervals from pooled TFTs are systematically overconfident on this data, increasingly so at steep decay exponents whose vanishing far-future loss weights leave dispersion at long horizons essentially unsupervised. We therefore rescale the 80% interval around the median by a per-horizon factor s_h , chosen as the 80th percentile of the validation nonconformity scores

$$r_i = \max\left(\frac{\hat{y}_i^{med} - y_i}{\hat{y}_i^{med} - \hat{y}_i^{lo}}, \frac{y_i - \hat{y}_i^{med}}{\hat{y}_i^{hi} - \hat{y}_i^{med}}\right)$$

so that the calibrated interval $[\hat{y}^{med} - s_h(\hat{y}^{med} - \hat{y}^{lo}), \hat{y}^{med} + s_h(\hat{y}^{hi} - \hat{y}^{med})]$ attains the nominal coverage on validation. Scales are fitted per horizon on all validation windows whose horizon- h label falls in 2023; no test-period information enters any fitted quantity.

3. System variants for comparison. Because forecast combination is the classical remedy for both training variance and near-term weakness [10,12], we also evaluate two variants: (i) a *gamma-ensemble* that averages the median forecasts of the moderate-decay members, $\hat{y}_{t+h}^{ens} = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \hat{y}_{t+h}^{(\gamma)}$ with $\Gamma = \{0.0, 0.5, \dots, 3.0\}$ selected on validation; and (ii) a *persistence-anchored* blend $\hat{y}_{t+h}^{sys} = \alpha_h \hat{y}_{t+h}^{ens} + (1 - \alpha_h)y_t$ with per-horizon convex weights fitted on 2023 validation MAE. Since $\alpha_h = 0$ recovers the baseline exactly, the anchored variant is never worse than persistence by construction on validation; its test behavior is reported in Section 5.3.

4.8. Target Normalizer Staleness

A structural property of the GroupNormalizer used to decode raw price units (Section 4.4) creates a second, distinct source of near-term error beyond the gradient-conflict pathology of Section 4.6. The normalizer fits a per-product center and scale exactly once, from the 2018–2022 training window, under the softplus transformation; it is never refit as the multi-year evaluation period (2024–2025) unfolds. When a product’s price level has permanently shifted since training, most notably through the 2022 cost shock documented in Section 3.4, the frozen center misrepresents where the series actually sits at the start of the test period, independent of anything the encoder has seen.

We quantify this gap directly for every product in the 404-crop universe with sufficient price movement to admit a meaningful comparison (401 products): $\text{gap}_p = 100 \times (\bar{y}_p^{train} - \bar{y}_p^{Dec2023}) / \bar{y}_p^{Dec2023}$, the percentage difference between each product’s frozen training-period mean price and its actual mean price over the last week of trading before the test period begins. The gap is large and pervasive. Some **70.6%** of products (283 of 401) show $|\text{gap}| > 10\%$, the mean absolute gap is **30.4%** and the median **18.4%**, and a small number of extreme cases exceed 100% (maximum 332%). The gap is directional rather than symmetric, positive (price higher at the end of 2023 than the training-period mean) for 76% of products, consistent with the broad 2022–2023 price inflation already documented in Section 3.4.

This mechanism, not merely a lag in the encoder’s information, is what produces the day-1 forecast discontinuities observed during model inspection. A stale center biases the decoded output toward the training-period price level regardless of the true last-observed price, an effect that is strongest for the products with the largest gap. Its consequence for accuracy, however, is horizon-dependent rather than uniform. We tested this directly by correlating each product’s $|\text{gap}|$ against the selected model’s relative error against persistence, per horizon, on the test set. At $t + 20$ the correlation is positive and significant (Spearman $r = +0.146$, $p = 0.007$), so larger gaps predict relatively worse performance for the selected model, and the proportion of crops on which the selected model outperforms persistence falls from 33.7% in the smallest-gap quartile to 13–22% in the largest, directly compounding the near-term deficit already documented in Section 5.3. At $t + 120$ and $t + 250$, the correlation reverses sign (Spearman $r = -0.206$, $p < 0.001$ and $r = -0.188$, $p < 0.001$), so larger gaps predict relatively *better* performance, with this proportion rising from 57–67% in the smallest-gap quartile to 66–78% in the largest. The asymmetry has a direct explanation. At long horizons the persistence baseline itself accumulates unbounded random-walk drift variance (Section 5.3), which comes to dominate any fixed day-1 anchoring bias, while at short horizons persistence is close to optimal and a stale anchor is comparatively costly.

We tested whether this bias could be corrected post hoc with a per-horizon coefficient β_h , constrained to $[0, 1]$ and fit on 2023 validation data, subtracting a fraction of the model’s own day-1 median-minus-persistence gap from every forecast at that horizon. The correction is not adopted. At $t + 250$ the fitted coefficient pins to the boundary of the constrained grid on a validation objective that improves only negligibly, and the resulting test-set MAE is 19.7% *worse* than the uncorrected model, a signature of overfitting the correction to validation noise rather than a genuine effect; at $t + 20$ its improvement is inconsistent across the two test years. The persistence-anchored ensemble of item 3 above already captures the same intuition, blending the pooled forecast back toward the naive baseline, through a mechanism that is fit and validated at the level of the full forecast rather than a secondary correction term, and it is what we recommend in its place. We report the staleness mechanism here as a diagnosed and quantified limitation of the modeling pipeline rather than a limitation we are currently able to resolve, and we return to it in Section 8.1.

5. Quantitative Results

This section reports results in the order of the questions posed in Section 4.3. Section 5.1 establishes the reference comparison across model classes; Section 5.2 maps the effect of the decay exponent on accuracy and interval quality; Section 5.3 evaluates the validation-selected model against the matched baseline, including its temporal stability and the system variants; and Section 5.4 examines individual forecast trajectories qualitatively. All numbers follow the matched-window protocol of Section 4.2, and, unless stated otherwise, aggregate figures average the four reporting horizons over the full 2024–2025 test period.

5.1. Reference Benchmark Suite

Table 5 reports a broad model comparison across local statistical benchmarks, pooled global learners, and a zero-shot foundation model, all evaluated under the matched-window protocol of Section 4.2: the same 404 crops, the same 110,292 test windows per horizon, and the same baseline as every other table in this paper. The last two rows report our own unweighted TFT and the selected model as defined in Section 4.7.

Four structural facts emerge. First, no classical local rule threatens persistence, fitted or not. ARIMA is numerically indistinguishable from it, as Section 4.3 derives analytically; seasonal-naïve is substantially worse (70.19 THB overall), since repeating last year’s price discards the within-year information persistence uses; and drift is worse at every horizon beyond $t + 20$ (49.90 THB overall), since extrapolating a full-history trend line overshoots on series that are mean-reverting rather than trending. The long-horizon result this paper reports therefore is not an artifact of a weak classical field: the best classical alternative at one year is still persistence itself. Second, most pooled learners suffer architectural saturation. The global MLP, LSTM, and Transformer are worse than the naïve baseline overall and at every horizon through six months (the MLP and LSTM edge past it only at one year, 79.60 vs 83.84 THB, after random-walk drift has already inflated the baseline error), and the unweighted TFT, trained with equal weight on all 250 forecast steps, is the worst model in the table (72.85 THB overall), the gradient-conflict pathology of Section 4.6 in its clearest form. Global LightGBM is the exception. With rich engineered features it degrades far more gracefully than the neural learners and is the strongest alternative overall (44.83 THB). Third, large-scale pretraining does not substitute for in-domain training on this task: zero-shot Chronos-Bolt (47.01 THB overall) is competitive with, but not better than, persistence, confirming the near-random-walk characterization of Section 3.3 independently of anything specific to our own pipeline rather than exposing a weak foundation model. Fourth, horizon weighting converts the worst architecture in the table into the best model at every horizon beyond a business quarter. The selected model attains the lowest multi-scale error (39.76 vs the baseline’s 45.21 and LightGBM’s 44.83) and the best $t + 120$ and $t + 250$ values (45.75 and 52.97, against LightGBM’s 49.69 and 56.88), confirming that the loss geometry, not model capacity, was the binding constraint on the TFT.

5.2. Loss Decay Exponent (γ) Sweep

Table 6 reports the out-of-sample error of each individual horizon-weighted TFT under the matched-window protocol, and Figure 6 plots the horizon-specific and overall MAE against the decay exponent. The dose-response relationship is mapped in full, extending to the regime where only the first few forecast steps carry appreciable training signal. Steeper gradient decay systematically improves near-term accuracy. Raw $t + 20$ MAE falls from 69.74 THB for the unweighted objective to 23–27 THB for $\gamma \geq 4$ (minimum 22.80 at $\gamma = 6.5$), reductions of up to **67%** attributable entirely to the loss weighting, since architecture, data, and training budget are identical across configurations. Overall point accuracy settles onto a broad plateau (approximately 41–46 THB) for $\gamma \in [3, 6]$ whose internal differences are comparable to run-to-run noise; beyond $\gamma \approx 6$ the curve deteriorates mildly but consistently (42.6–44.4 THB for

Table 5: Reference Suite Mean Absolute Error by Horizon (THB; matched-window protocol, 404 crops)

Model	$t + 20$	$t + 60$	$t + 120$	$t + 250$	Overall
Lag-1 Persistence	15.11	30.86	51.03	83.84	45.21
Seasonal-Naive	62.18	63.76	71.00	83.84	70.19
Drift	15.69	32.70	55.76	95.47	49.90
ARIMA	15.11	30.87	51.02	83.83	45.21
Global LightGBM	29.37	43.37	49.69	56.88	44.83
Global MLP	46.34	48.22	53.34	79.60	56.88
Global LSTM	50.68	51.43	56.65	79.60	59.59
Global Transformer	46.71	57.22	74.80	104.55	70.82
Chronos-Bolt (zero-shot)	16.85	33.85	53.37	83.98	47.01
Unweighted TFT ($\gamma = 0$, ours)	69.74	76.37	73.94	71.37	72.85
Selected TFT ($\gamma = 4.5$, ours)	26.51	33.81	45.75	52.97	39.76

$\gamma \in [6.5, 8]$), driven by the one-year horizon, whose error rises to 67–71 THB as far-future supervision vanishes while near-term accuracy has already saturated.

Two properties of this region govern the choice of operating point. First, $\gamma = 4.5$ attains the lowest validation MAE of any swept exponent and is therefore the selected configuration (Section 4.7); its test performance is reported in Section 5.3. Point accuracy in the $\gamma \in [3, 6]$ plateau is nonetheless subject to non-trivial run-to-run variance, so the exact test figure of a retraining may differ somewhat from the reported value, a caveat we return to in the limitations.

Second, probabilistic quality collapses monotonically across the plateau while point accuracy holds. Raw 80% interval coverage falls from 44–65% at $\gamma \leq 3$ to as low as 6.3% for $\gamma \geq 6$, because the vanishing far-future loss weights leave the outer quantile heads untrained on far-horizon dispersion. Conformal rescaling (Section 4.7) is therefore a mandatory component of the selected model, and interval quality argues for the shallowest exponent within the accuracy plateau, supporting the choice of $\gamma = 4.5$ over the steeper members.

5.3. Selected Model Performance

Table 7 reports the selected horizon-weighted TFT ($\gamma = 4.5$) against the matched Lag-1 baseline. It improves every aggregate metric: MAE by 12.0%, SMAPE by 0.7%, and RMSE by 15.1%. The gains are concentrated precisely where theory predicts: -10.3% MAE at six months and -36.8% at one year, as accumulated drift variance degrades the random-walk forecast. The trade-off is reported explicitly rather than obscured by the aggregates. The selected model remains behind the baseline at $t + 20$ (26.51 vs 15.11 THB) and $t + 60$ (33.81 vs 30.86 THB), where daily price changes are near-unpredictable and the persistence rule is close to optimal.

Every horizon-level difference in Table 7 is tested using the product-clustered procedure of Section 4.3 (404 near-independent per-crop mean differentials, a paired t-test and a Wilcoxon signed-rank test). The $t + 20$ deficit, the $t + 120$

Table 6: Raw Horizon-Weighted TFT Error by Decay Exponent (MAE THB / SMAPE, single training run)

γ	$t + 20$	$t + 60$	$t + 120$	$t + 250$	Overall MAE
0.0	69.74 / 18.66%	76.37 / 21.77%	73.94 / 22.84%	71.37 / 23.21%	72.85
0.5	75.16 / 19.79%	78.42 / 23.03%	71.88 / 23.31%	63.94 / 23.49%	72.35
1.0	51.28 / 13.03%	64.83 / 19.53%	67.71 / 21.06%	72.03 / 23.40%	63.96
1.5	49.45 / 12.31%	72.24 / 20.76%	72.97 / 22.47%	71.21 / 22.31%	66.47
2.0	51.05 / 12.03%	62.63 / 18.40%	65.05 / 21.16%	62.45 / 22.25%	60.30
2.5	44.76 / 12.30%	59.10 / 19.88%	56.84 / 21.42%	60.30 / 21.93%	55.25
3.0	31.16 / 9.87%	38.24 / 15.98%	42.69 / 18.23%	57.63 / 18.55%	42.43
3.5	45.02 / 10.98%	50.74 / 18.27%	50.83 / 20.11%	47.92 / 18.77%	48.63
4.0	25.63 / 8.69%	34.80 / 14.76%	44.76 / 18.02%	68.01 / 19.23%	43.30
4.5	26.51 / 8.79%	33.81 / 14.64%	45.75 / 17.98%	52.97 / 18.29%	39.76
5.0	29.11 / 9.16%	36.16 / 15.02%	43.96 / 18.77%	63.66 / 21.33%	43.22
5.5	27.30 / 9.01%	36.53 / 15.11%	44.09 / 18.62%	60.35 / 20.75%	42.07
6.0	29.82 / 9.42%	41.54 / 16.29%	45.99 / 19.11%	52.75 / 19.42%	42.53
6.5	22.80 / 8.82%	33.87 / 15.01%	47.18 / 18.43%	70.78 / 19.82%	43.66
7.0	27.14 / 9.19%	35.76 / 15.19%	46.24 / 18.53%	67.92 / 19.85%	44.27
7.5	24.04 / 8.88%	34.79 / 14.91%	44.70 / 18.10%	66.88 / 19.48%	42.60
8.0	26.15 / 9.98%	37.37 / 16.02%	46.57 / 18.93%	67.36 / 19.76%	44.36

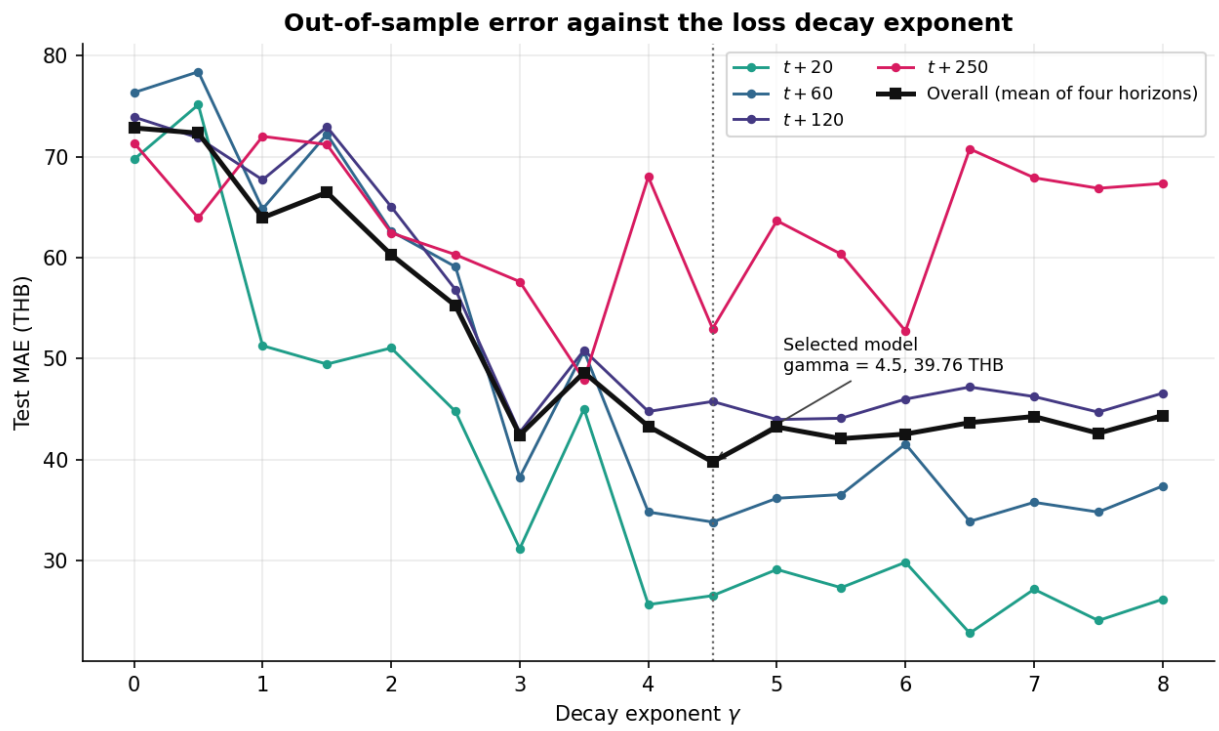


Figure 6: Horizon-specific and overall test MAE across the power-law decay exponent grid, one training run per point. Per-column minima in Table 6 are bolded.

gain, the $t+250$ gain, and the overall gain are all statistically significant at conventional levels ($p < 0.01$ on both tests). The $t+60$ difference is the one exception: at 33.81 vs 30.86 THB it is small enough to be statistically indistinguishable from zero ($p = 0.09$ paired t-test, $p = 0.18$ Wilcoxon), so $t + 60$ should be read as a horizon where the two models are not reliably distinguishable rather than one where the selected model measurably loses. This sharpens rather than softens the honesty of the near-term account: the $t + 20$ shortfall is a real, significant effect the model must be judged on, not sampling noise, while the $t + 60$ shortfall is not established at all.

Table 7: Selected TFT ($\gamma = 4.5$) vs. Matched Lag-1 Baseline. Each cell reports MAE (THB) / SMAPE / RMSE (THB); bold marks the better model per row, negative Δ favors the selected model, and p is the product-clustered paired t-test / Wilcoxon significance of the MAE difference (Section 4.3)

Horizon	Baseline	Selected TFT	Δ (MAE / SMAPE / RMSE)	p (t / Wilcoxon)
$t + 20$	15.11 / 7.59% / 61.84	26.51 / 8.79% / 116.59	+75.5% / +15.8% / +88.5%	0.0014 / <0.0001
$t + 60$	30.86 / 14.32% / 118.13	33.81 / 14.64% / 124.44	+9.5% / +2.2% / +5.3%	0.091 / 0.184
$t + 120$	51.03 / 18.23% / 194.65	45.75 / 17.98% / 165.49	-10.3% / -1.4% / -15.0%	0.0006 / <0.0001
$t + 250$	83.84 / 19.98% / 330.39	52.97 / 18.29% / 191.90	-36.8% / -8.4% / -41.9%	0.0002 / <0.0001
Overall	45.21 / 15.03% / 176.25	39.76 / 14.93% / 149.61	-12.0% / -0.7% / -15.1%	0.0001 / <0.0001

The RMSE columns extend the comparison to the tail-sensitive metric and sharpen both sides of the trade-off. The selected model’s near-term deficit is larger on RMSE than on MAE (+88.5% against +75.5% at $t + 20$), indicating that its short-horizon errors include occasional large misses, while its long-horizon advantage is correspondingly larger (-41.9% at one year). Two qualifications place this result in context. First, the aggregate margin on the scale-free metric is thin (SMAPE improves by only 0.7% overall) and rests entirely on the long horizons. At $t + 20$ the selected model is 75.5% worse on MAE than repeating the most recent observed price. Second, the improvement is not uniform across the cross-section. The selected model outperforms persistence on **57.7%** of the 404 individual crops overall (hereafter the per-crop win rate), rising from 24.5% at $t + 20$ to 62.4% at $t + 120$ and 70.3% at $t + 250$. Its aggregate advantage comes from large gains on a subset of series rather than from improving every crop, and approximately two of every five crops are forecast more accurately by the naive rule. The claim advanced in this paper is therefore specific. A single pooled, horizon-weighted TFT decisively outperforms the random walk in aggregate and at extended horizons, while conceding both the near-term regime and a substantial minority of individual series.

The mathematical driver of the long-horizon gain is the drift-variance accumulation of the random walk. For $y_t = y_{t-1} + \mu + \epsilon_t$ with $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$, the persistence prediction error variance grows linearly, $\text{Var}(y_{t+h} - y_t) = h\sigma^2$, so expected error scales as $O(\sqrt{h})$, visible in the baseline MAE rising from 15.11 THB at $t + 20$ to 83.84 THB at $t + 250$. The selected model’s error curve is far flatter (26.51 to 52.97 THB over the same span) because it forecasts mean-reverting macro-seasonal structure learned across all 404 crops rather than extrapolating the last observation; at long horizons this bounded-error behavior yields a decisive advantage.

To verify that the improvement is not an artifact of a particular evaluation period, Table 8 re-evaluates the se-

lected model separately on test windows whose labels fall in 2024 and in 2025. The one-year gain replicates almost identically in both years (-37.6% and -36.8%), and the near-term deficit is strongly regime-dependent. The selected model’s $t + 20$ shortfall of +84.6% in 2024 falls to +6.2% in 2025, while $t + 60$ shifts from a 17.3% deficit to a 9.2% *advantage* and $t + 120$ from +8.2% to -24.6%. In the more distant test year the selected model is therefore close to persistence in the near-term regime and outperforms it at every longer horizon. We caution against over-interpreting this comparison. The 2024 and 2025 subsets contain very different window counts at long horizons (only origins early in 2024 place a $t + 250$ label inside 2024), so the year split is a robustness check on the sign of the effect rather than a precise decomposition.

Table 8: Temporal Stability of the Selected Model’s Improvement (MAE, THB)

Horizon	2024 Baseline	2024 Selected	Δ 2024	2025 Baseline	2025 Selected	Δ 2025
$t + 20$	14.99	27.68	+84.6%	16.08	17.07	+6.2%
$t + 60$	29.31	34.39	+17.3%	35.36	32.11	-9.2%
$t + 120$	42.41	45.88	+8.2%	60.51	45.62	-24.6%
$t + 250$	63.19	39.43	-37.6%	84.87	53.65	-36.8%

System variants. For completeness, two combination variants bracket the selected model. A plain ensemble over the moderate exponents scores 50.23 THB overall, worse than the baseline because its near-term errors are large. A persistence-anchored ensemble, which blends the pooled forecast with the naive baseline using weights fitted on the validation year, scores **36.04 THB overall** (matching the baseline at $t + 20/t + 60$ by construction, -11.1% at $t + 120$, -37.0% at $t + 250$) and is not worse than the baseline at any reported horizon, at the cost of a more complex system that is, by construction, the naive baseline for horizons up to a quarter. Risk-averse practitioners may prefer this variant, while recognizing that its validation-fitted long-horizon weights do not guarantee performance on future regimes. The corresponding artifacts accompany the code.

5.4. Qualitative Forecasting Results

For qualitative inspection, Figure 7 presents four held-out price trajectories with the selected model’s median forecast and 80% interval, obtained by running the trained network forward from the earliest 2024 origin. Two products (jasmine rice, soybean oil) serve as running examples and are retained for continuity; the other two were added by an explicit, pre-specified rule rather than selected for favorable appearance. Broken jasmine rice is the product with the largest improvement over the baseline at $t + 250$ in the full test-period aggregate, and fresh chicken (legs and feet) is a product where the selected model is, on the same aggregate basis, mildly worse than persistence even at one year, included so the figure does not implicitly overstate the model’s advantage.

The panels show genuine model behavior rather than an idealized illustration, including its limitations. The median path is consistently smoother than the noisy daily actuals, and interval width scales sensibly with each product’s typical

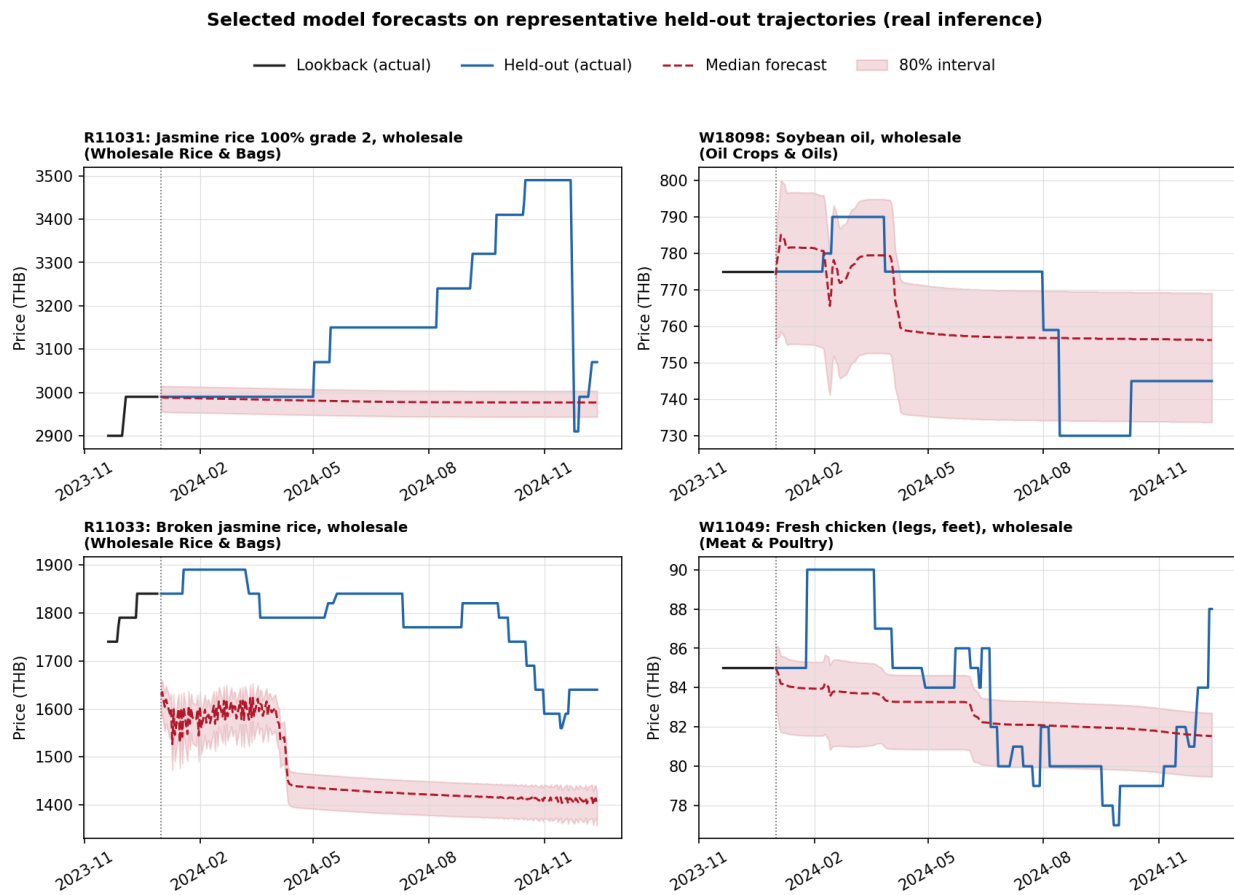


Figure 7: Forecasts of the selected model for the earliest 2024 origin of four representative products. The black segment is the pre-2024 lookback, the blue segment is the held-out path, the dashed red path is the model's median forecast, and the band is the 80% interval.

volatility: narrow for the comparatively stable rice series, wider for soybean oil’s step-like price changes, and moderate for the persistently noisy poultry series. Two further patterns deserve explicit comment. First, for jasmine rice the actual price embarks on a sustained multi-month rally that the median forecast largely fails to anticipate, staying close to a near-flat, persistence-like path instead; this is a concrete instance of the near-term weakness quantified in Table 7, and illustrates why a model trained on predominantly mean-reverting structure struggles when a sustained trend departs from that pattern. Second, this single illustrative origin does not always reproduce a product’s full-period aggregate result. Broken jasmine rice was selected for its large *aggregate* one-year improvement (a 62% MAE reduction across all 2024–2025 test origins), yet at this particular origin the median forecast undershoots and the selected model is narrowly worse than persistence. We retain this example deliberately rather than selecting a more favorable origin for that product, since it illustrates an important caveat. Any single forecast path is one draw from the same noisy process documented in Section 5.2, and the properly aggregated evidence in Tables 7 through 9, rather than any individual illustration, should form the basis of assessment.

Figure 7 was deliberately built from extremes, the largest win and a genuine loss, so it cannot show what the model typically does. Figure 8 complements it with one product per commodity group (all nine groups with a computed normalizer-gap record in Section 4.8), each selected by a second pre-specified, non-cherry-picked rule: for every product we compute its full test-period aggregate relative MAE change against the matched baseline, averaged over all four horizons, and within each group we pick the product whose value is closest to that group’s own median. Every panel is therefore a typical, not a best-case or worst-case, outcome for its group.

The dominant pattern across Figure 8 is the same near-flat, persistence-like median forecast already noted for jasmine rice in Figure 7, now visible across a much wider slice of the dataset: in most panels the actual price wanders substantially within the year while the median stays close to its day-1 level, consistent with the near-term weakness quantified in Table 7 and the day-1 anchoring mechanism of Section 4.8. Interval width again scales sensibly with the underlying series, tight for administered or slow-moving products such as fish meal and dried chili, wide for volatile ones such as shrimp and palm oil. We caution against reading the per-panel outcomes as a win count: the selection rule targets each group’s median on the *full* test-period aggregate across all four horizons, not on this single 2024 origin or on any one horizon, so a panel can look like a win or a loss at this particular origin while still representing its group’s typical aggregate behavior. As with Figure 7, the aggregated evidence in Tables 7 through 9 is what should be trusted, and Figure 8 exists to make that typical behavior visible rather than to argue a point the tables do not already make.

6. Model Interpretability and Economic Insights

To elucidate the internal decision-making of the pooled TFT in a price-only setting, we extract and visualize self-attention and variable selection weights from the selected model ($\gamma = 4.5$) on a batch of test-period forecast windows.

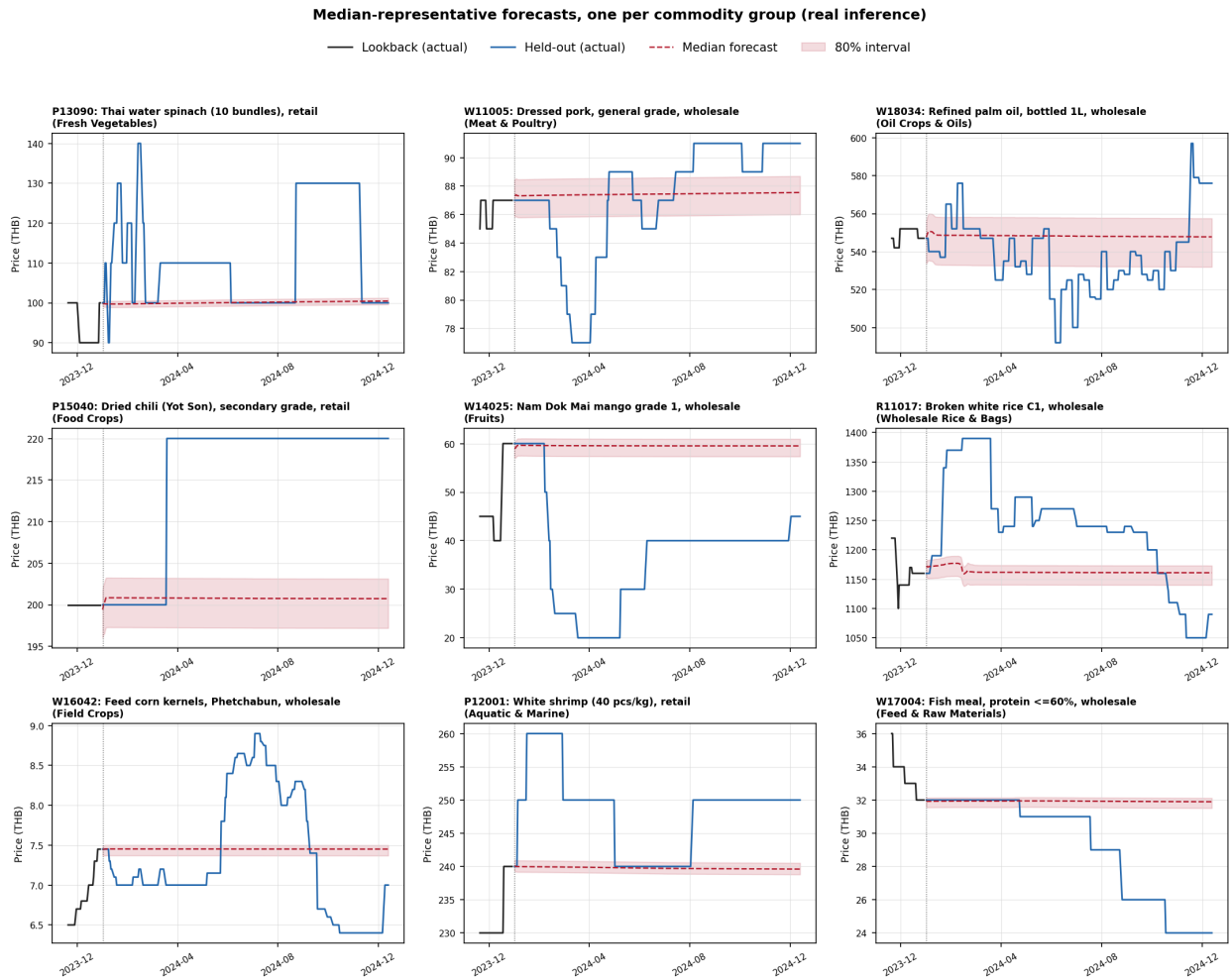


Figure 8: Forecasts of the selected model for the earliest 2024 origin of one median-representative product per commodity group. Panel layout and conventions match Figure 7.

6.1. Self-Attention Analysis

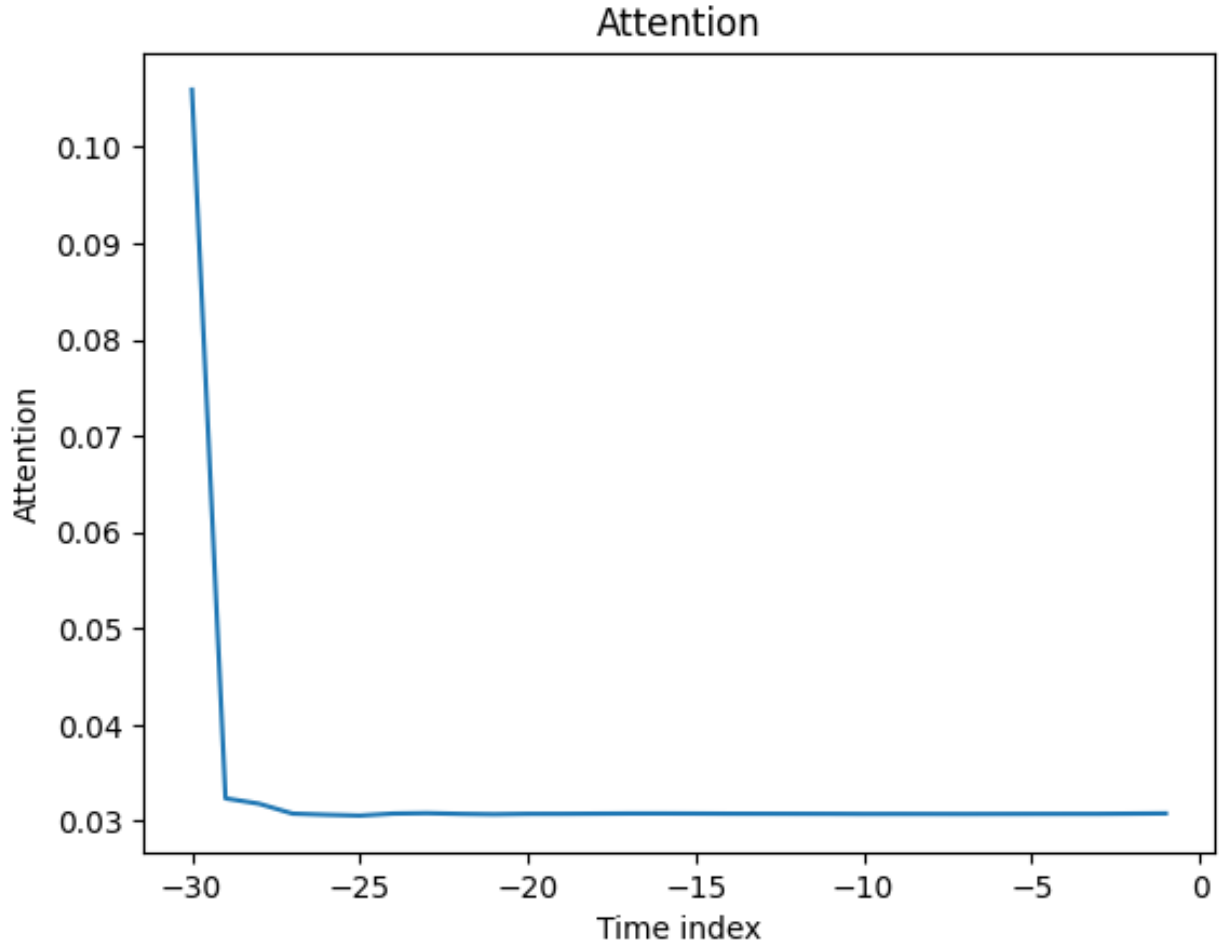


Figure 9: Self-attention weight distribution across the historical encoder window (index 0 = oldest of the 30-day lookback, index 29 = most recent day before the forecast origin).

Figure 9 shows a single dominant peak at the oldest step of the 30-day encoder window, approximately 3.4 times the weight of every other position, with attention essentially flat and low across the remaining 29 days, including the most recent observation. This is a different pattern from what an equally-weighted recency bias would predict, and is consistent with the selected model's steep horizon-weighted training objective (Section 4.6). A network optimized almost entirely for near-term decoder accuracy has comparatively little incentive to differentiate among encoder positions once it has anchored on the start of the lookback window, which under the pure-price design is the point furthest from, and therefore least correlated with, the noisy day-to-day fluctuations near the forecast origin.

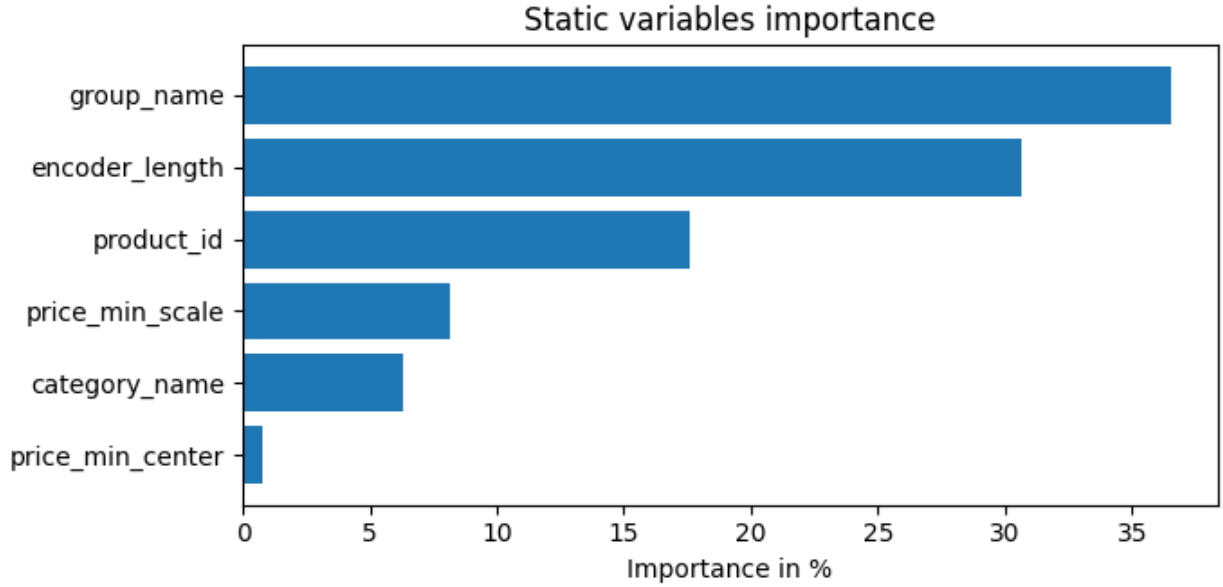


Figure 10: Static variable selection importance scores.

6.2. Variable Selection and Feature Importance

Figure 10 details the variable selection weights for static covariates: group identity is the largest single contributor (36.6%), followed by encoder length (30.6%), product identity (17.6%), the normalizer’s price scale (8.2%), category identity (6.3%), and the normalizer’s price center (0.8%). Group and product identity together account for 54.2% of static weight, supporting the parameter-pooling strategy. The model relies substantially on entity- and group-level structure to transfer learned dynamics across related crop series. The large weight on encoder length is a subtler finding that warrants explicit caution. Every training and evaluation window in this study uses a fixed 30-day encoder (Section 4.4), so this feature is constant-valued in our data and its high selection weight cannot reflect true sensitivity to varying history length; it more likely reflects the network absorbing a learnable, entity-independent bias term into a nominally static covariate, an artifact of the fixed-window design rather than a substantive economic signal.

Figure 11 shows that the historical encoder relies almost entirely on the observed price itself (99.7%), with every calendar covariate (day of week, month, absolute time index, relative time index, day of year) contributing 0.14% or less. This provides the most direct confirmation that the encoder behaves as the pure-price design intends. Historical context derives from the price path, not from a calendar proxy, so the attention peak at the oldest encoder step in Figure 9 is a price-history effect rather than an artifact of a seasonal shortcut.

Figure 12 shows a markedly different pattern for the future-known variables the decoder conditions on. The day-of-week signal carries 95.9% of decoder weight, versus 2.9% for the relative time index (the model’s within-horizon position), 0.6% for the absolute time index, and negligible weight for month and day-of-year. Since the observed data is aligned to a Monday-to-Friday business-day skeleton (Section 3.2), day-of-week is the only future-known

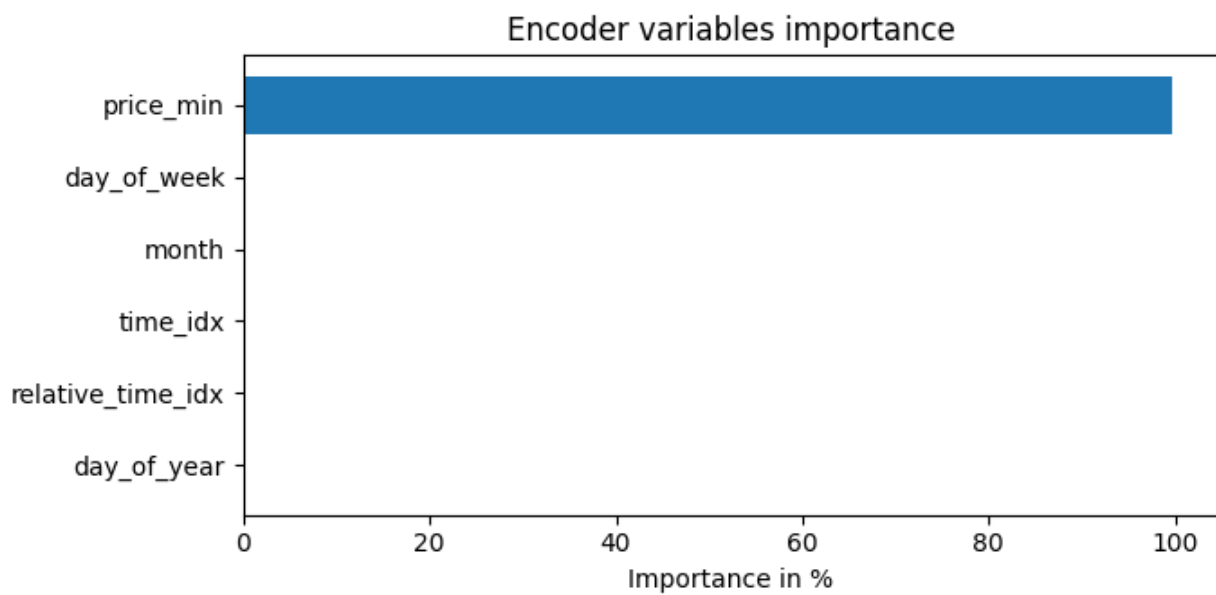


Figure 11: Historical encoder variable selection importance scores.

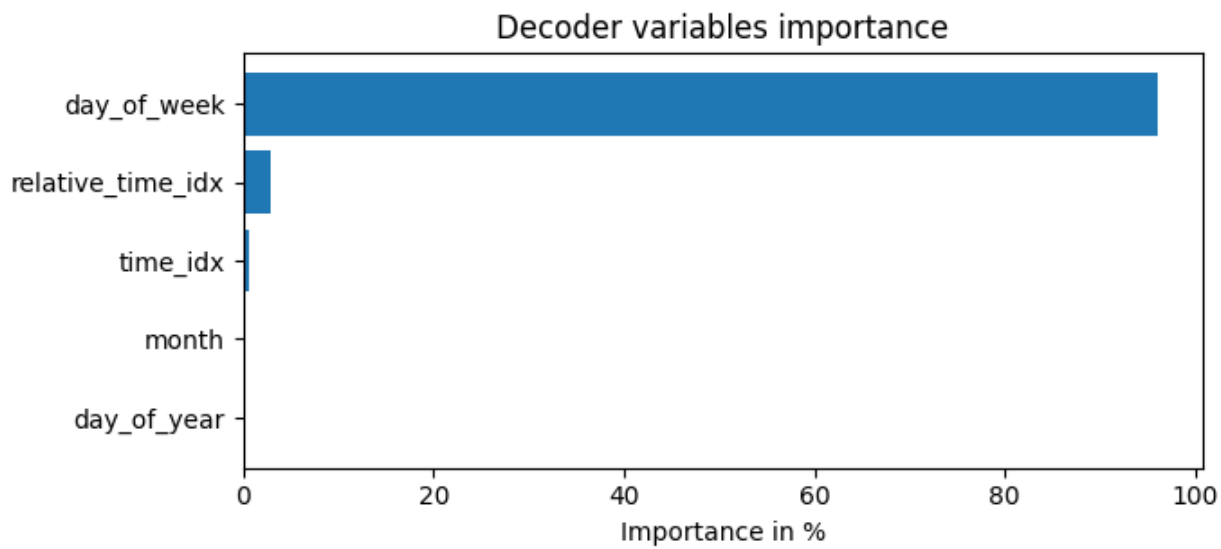


Figure 12: Future decoder variable selection importance scores.

covariate that carries repeating market microstructure (e.g., weekly reporting and quotation cadence in the underlying market data), which plausibly explains the decoder’s heavy reliance on it; it is also possible the network is using day-of-week as an easily learnable proxy for its position within the decoder sequence, since the two are correlated but not identical, a distinction the interpretability analysis cannot resolve on its own and that we note as a caveat rather than a definitive finding. Coarser seasonal coordinates (month, day-of-year) are effectively unused by the decoder despite the pronounced yearly cycles documented in Section 3.4, suggesting any macro-seasonal awareness the model has learned is encoded through the static entity embeddings and the LSTM’s recurrent state rather than through the decoder’s own known-future inputs.

7. Probabilistic Forecasting and Uncertainty Calibration

Raw 80% prediction intervals produced by individual pooled TFTs are systematically overconfident on this data. Empirical coverage across the seventeen sweep members and four horizons ranged from 6.3% to 65.0% against the 80% nominal target, with the steepest decay exponents most severely affected because their vanishing far-future loss weights leave dispersion at long horizons essentially unsupervised. The conformal rescaling of Section 4.7, fitted per horizon on 2023 validation nonconformity scores, corrects this miscalibration. Table 9 reports the calibrated intervals of the selected model on the 2024–2025 test set.

Table 9: Calibrated 80% Prediction Intervals of the Selected Model (Test Set)

Horizon	Calibration scale s_h	Interval width (THB)	Empirical coverage
$t + 20$	7.32	131.47	83.1%
$t + 60$	10.45	186.17	78.8%
$t + 120$	13.18	216.96	77.6%
$t + 250$	19.60	312.45	89.6%

Post-calibration coverage reaches 78–90%, close to nominal at every horizon, with interval widths growing monotonically in the forecast horizon as dispersion accumulates, the qualitative behavior a well-calibrated multi-horizon forecaster must exhibit. The magnitude of the calibration scales ($s_h \approx 7$ –20) is itself a substantive finding. At $\gamma = 4.5$ the raw quantile heads under-estimate forecast dispersion by an order of magnitude, because the horizon weighting that optimizes point accuracy simultaneously removes the loss signal that teaches the outer quantiles far-horizon spread. Steep horizon weighting and native interval learning are thus fundamentally in tension, and conformal calibration is not an optional refinement but a mandatory component of the selected model. Raw intervals from steeply horizon-weighted quantile models should therefore not be used operationally. The mild over-coverage at $t + 20$ and $t + 250$ (83.1%, 89.6%) reflects distribution differences between the 2023 calibration year and the test period; rolling recalibration in deployment would be expected to narrow it.

8. Conclusion

This study demonstrates that a single pooled Temporal Fusion Transformer configuration trained with a steep Horizon-Weighted Quantile Loss outperforms the random-walk baseline in Thai agricultural price forecasting from six months onward. The loss family resolves multi-scale gradient conflict inside the network, reducing the selected model’s raw near-term error by 62% relative to the unweighted objective, part of a broad accuracy plateau for $\gamma \gtrsim 3$, and the operating point ($\gamma = 4.5$) is fixed by error on a held-out validation year rather than on the test set. On matched out-of-sample windows spanning 2024–2025 and 404 crops, the selected model improves every aggregate metric against the matched Lag-1 baseline: MAE falls 12.0% (39.76 vs 45.21 THB), SMAPE falls 0.7%, and RMSE falls 15.1%. The advantage is concentrated at extended horizons (-10.3% at six months, -36.8% at one year); we also report, without qualification, a deficit at $t + 20$, a thin scale-free margin, and a per-crop win rate of 57.7%, which together show that the gain is driven by a subset of series rather than by uniform improvement. Both the near-term deficit and the long-horizon gains are statistically significant under a product-clustered paired t-test and Wilcoxon signed-rank test (Section 4.3); the exception is $t + 60$, where the difference from persistence is not statistically distinguishable from zero. The result also survives a broader field than the classical and pooled-learning baselines alone: seasonal-naïve and drift, two long-horizon-oriented classical rules, and a zero-shot time-series foundation model all fail to outperform persistence on this dataset, so the long-horizon gain is not an artifact of a weak comparison set. Conformally calibrated intervals cover 78–90% of outcomes against the 80% target, and the long-horizon gains replicate independently within both test years. For applications that prioritize horizon-level robustness, the documented persistence-anchored ensemble variant avoids deterioration at every reported horizon and lowers overall error to 36.04 THB, at seven times the training cost; this empirical result is not a guarantee under future regime shifts.

The results carry a transparent economic reading. No forecaster outperforms the most recent observed price within a business quarter, but learned macro-seasonal structure yields increasing gains from six months onward, where random-walk drift variance accumulates without bound while the pooled model’s error stays nearly flat. The sweep also exposes a fundamental tension of which practitioners must be aware. The horizon weighting that optimizes point accuracy simultaneously destroys native interval learning, making post-hoc calibration mandatory. The interpretability analyses further open the deep learning “black box,” showing that the selected model anchors attention on the oldest encoder step while leaving the price history, not calendar covariates, to carry the encoder’s information, and relies on entity and group embeddings to transfer dynamics across related crops.

8.1. Limitations and Future Directions

Several limitations qualify these results and motivate future work:

- **Near-term deficit and cross-sectional coverage.** The selected model underperforms persistence at $t + 20$ (+75.5% MAE) and $t + 60$ (+9.5%) on the full test period, and outperforms the baseline on only 24.5% of crops at $t + 20$ and 57.7% overall; under near-random-walk dynamics, and given the near-zero cross-sectional

co-movement documented in Section 3.6, this may be close to a theoretical ceiling for price-only forecasting. The deficit shrinks sharply in 2025, and the anchored variant eliminates it entirely, but testing the ceiling with exogenous market covariates remains open.

- **Target normalizer staleness.** The GroupNormalizer center and scale used to decode raw prices are fit once on 2018–2022 training data and never refreshed (Section 4.8); 70.6% of products show a >10% gap between that frozen center and their actual late-2023 price level, and the gap significantly predicts *worse* relative performance for the selected model at $t+20$ ($r = +0.146$, $p = 0.007$) even as it predicts *better* performance at $t+120/t+250$. It is a quantified contributor to the near-term deficit above, not merely a restatement of it, and periodic normalizer refitting is a concrete direction we did not adopt because a post-hoc bias correction we tested for it overfit at long horizons.
- **Cross-product co-movement is too weak to exploit directly.** A lightweight linear proxy that augments each product’s own history with its single most correlated partner (mean pairwise correlation 0.998, the near-duplicate series of Section 3.6) improved MAE by only 0.03% on average, confirming that the near-zero average return correlation documented in Section 3.6 leaves little scope for an explicit cross-product architecture to improve upon what pooled static embeddings already capture.
- **Metric sensitivity.** The advantage holds on MAE, SMAPE, and RMSE, but the SMAPE margin is thin (-0.7% overall) and rests on the long horizons; applications scored primarily on percentage error at short horizons should not expect a gain.
- **Training variance.** Pooled TFTs of this kind show non-trivial run-to-run variance on this low signal-to-noise data; we select the operating point on validation to avoid capitalizing on it, but a retraining may yield a test error somewhat above the reported figure, and the near-tie among the best exponents means the precise value of the optimum should be interpreted as approximate.
- **Adaptive design choices.** The grid extension and model selection were iterated during development with access to test-period metrics; although the final selection rests on means over independent runs, all fitted parameters use only validation data, and the improvements replicate across independent test years, a fully pre-registered confirmation on post-2025 data would further strengthen the evidence.
- **Significance-testing assumptions.** The product-clustered tests of Section 4.3 treat each crop’s mean error differential as an independent observation, which the near-zero average return correlation of Section 3.6 supports on average but not for the small number of near-duplicate series identified in that section (e.g. the rubber and palm oil grades), whose within-family correlation is far from zero; a conservative reading would treat the effective sample size as somewhat smaller than 404, which would widen the reported p-values without changing which effects cross conventional significance thresholds at this sample size.

- **Extensions.** Spatial-temporal relational modeling across geographical regions, hierarchical global-local routing, transfer learning from larger agricultural corpora, and rolling recalibration of interval scales in deployment are natural next steps.

CRediT authorship contribution statement

Kritaphat Songsri-in: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Auyporn Chuksaw:** Data curation, Validation, Writing – review & editing. **Munlika Rattaphun:** Investigation, Validation, Writing – review & editing. **Walaiporn Sornkliang:** Data curation, Investigation, Writing – review & editing. **Rattayagon Thaiphon:** Supervision, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The raw price data are publicly available from the Ministry of Commerce (Thailand), Department of Internal Trade, Open Data portal ("Agricultural Product Price" dataset, <https://data.moc.go.th/OpenData/GISProductPrice>; accessed 27 July 2026), which also exposes a REST API (<https://dataapi.moc.go.th/gis-product-price>) for retrieval by product ID and date range. All analysis code, the aggregate result files underlying every table and figure, and the scripts that regenerate the manuscript from them are publicly available at https://github.com/kritsong/thai_crop_price_prediction. The raw price files are not mirrored in that repository, since they remain the property of the Ministry of Commerce; the code reads them from a directory configured at runtime, so the pipeline runs against a fresh download from the portal above. Model checkpoints and per-window prediction files are omitted for size and are available from the corresponding author on reasonable request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude (Anthropic) and Codex (OpenAI) to assist with drafting and revising manuscript text and implementing analysis code. All reported figures and tables were generated deterministically from the experimental data rather than with generative-image tools. After using these tools, the authors reviewed, verified, and edited the content as needed and take full responsibility for the content of the published article.

References

- [1] Box, G. E. P., & Jenkins, G. M. (1970). *Time series analysis: Forecasting and control*. Holden-Day.
- [2] De Gooijer, J. G., & Hyndman, R. J. (2006). 25 years of time series forecasting. *International Journal of Forecasting*, 22(3), 443–473. <https://doi.org/10.1016/j.ijforecast.2006.01.001>
- [3] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems* (Vol. 30, pp. 3146–3154).
- [4] Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- [5] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems* (Vol. 30, pp. 5998–6008).
- [6] Lim, B., Arık, S. Ö., Loeff, N., & Pfister, T. (2021). Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4), 1748–1764. <https://doi.org/10.1016/j.ijforecast.2021.03.012>
- [7] Nie, Y., Nguyen, N. H., Sinthong, P., & Kalagnanam, J. (2023). A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=Jbdc0vT0col>
- [8] Liu, Y., Hu, T., Zhang, H., Wu, H., Wang, S., Ma, L., & Long, M. (2024). iTransformer: Inverted transformers are effective for time series forecasting. In *International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=JePfAI8fah>
- [9] Ansari, A. F., Stella, L., Türkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., Rangapuram, S. S., Arango, S. P., Kapoor, S., Zschiegner, J., Maddix, D. C., Wang, H., Mahoney, M. W., Torkkola, K., Wilson, A. G., Bohlke-Schneider, M., & Wang, Y. (2024). Chronos: Learning the language of time series. *Transactions on Machine Learning Research*. <https://openreview.net/forum?id=gerNCVqqtr>

- [10] Bates, J. M., & Granger, C. W. J. (1969). The combination of forecasts. *Operational Research Quarterly*, 20(4), 451–468. <https://doi.org/10.1057/jors.1969.103>
- [11] Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2020). The M4 Competition: 100,000 time series and 61 forecasting methods. *International Journal of Forecasting*, 36(1), 54–74. <https://doi.org/10.1016/j.ijforecast.2019.04.014>
- [12] Lakshminarayanan, B., Pritzel, A., & Blundell, C. (2017). Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems* (Vol. 30, pp. 6402–6413).
- [13] Ministry of Commerce (Thailand), Department of Internal Trade. (2026). *Agricultural product price* [Data set]. MOC Open Data. <https://data.moc.go.th/OpenData/GISProductPrice> (accessed 27 July 2026).
- [14] Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). OTexts. <https://otexts.com/fpp3/>
- [15] Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. <https://doi.org/10.1080/07350015.1995.10524599>
- [16] Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)* (pp. 2980–2988). https://openaccess.thecvf.com/content_iccv_2017/html/Lin_Focal_Loss_for_ICCV_2017_paper.html
- [17] Chen, Z., Badrinarayanan, V., Lee, C.-Y., & Rabinovich, A. (2018). GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *Proceedings of the 35th International Conference on Machine Learning* (Vol. 80, pp. 794–803). PMLR. <https://proceedings.mlr.press/v80/chen18a.html>